MarketSage AI

Global Event-Driven Real-Time Market Prediction & Options Trading Intelligence System

Master Engineering Blueprint, Research Strategy and Phase-Gated Execution Plan

Duniya mein kahin bhi kuch hota hai — Trump bolta hai, OPEC cut karta hai, RBI rate badalta hai — system 1 second ke andar us event ko detect kare, Nifty/BankNifty par uska impact nikale, aage ki candles ka distribution *predict kare, aur ek complete Options trade plan de — live market mein, bina kisi human ke.*

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This document is an engineering and research blueprint. It is not investment advice. Every performance number stated is a *design target subject to falsification*, not a promise. Chapter 3 exists specifically to destroy optimistic assumptions before they cost money.

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CHAPTER 1

Executive Summary

1.1 One-paragraph definition

MarketSage AI is a **distributional, event-conditioned forecasting system** for Indian index derivatives. It continuously ingests global news and market microstructure, detects market-moving events within milliseconds, estimates the *probability distribution* of Nifty/BankNifty returns over the next 1–30 minutes conditioned on that event plus current market state, converts that distribution into an optimal options structure using a fitted implied volatility surface, sizes the position under a strict risk budget, and continuously scores its own calibration in public.

1.2 What changed versus the original concept

The original vision is directionally correct but rests on six assumptions that will silently destroy the system in a live market. This document keeps the vision and replaces those assumptions with defensible engineering.

Original assumption	Why it breaks	What we build instead
Predict exact future OHLC (ghost candles)	Point forecasts of price are close to unforecastable; a single predicted candle is false precision and invites oversizing	**Fan-chart candles**: predicted quantile bands ($q_{05} \dots q_{95}$) from quantile regression plus a path sampler. Honest, and visually far more impressive
60–70% directional accuracy	Any honest out-of-sample 5-minute directional model lands at 52–57%. Higher on paper almost always means leakage	Optimise **expected value after costs**, not hit-rate. Report calibration (Brier score, reliability curve), not accuracy alone
Buy naked Call/Put after a spike	By the time the event is public, implied volatility has already expanded. Direction can be right and the option still loses (**IV crush**)	Structure selection engine: naked / vertical debit / ratio / straddle chosen by comparing forecast move against **implied move**
Sub-second latency beats institutions	You will never beat co-located HFT on wire speed. That is not the game	Edge is **interpretation speed and breadth**, not raw latency. Target window: seconds-to-minutes drift, not microsecond arbitrage
News API leads straight to a trade	Fake and erroneous headlines exist. The 2013 AP Twitter hack erased roughly USD 136 billion of market cap in three minutes	**Credibility-weighted multi-source confirmation**, severity gating and hard kill-switches. Auto-execution only after three months of a proven paper record

Original assumption	Why it breaks	What we build instead
Backtest on downloaded candles	Without a point-in-time news archive and historical option chains the backtest is fiction	**Point-in-time event store** (publish timestamp, never ingest timestamp) plus purchased 1-minute historical option chain. This is the single largest line item in the budget

1.3 The thesis in one line

Core Alpha Hypothesis

Retail and mid-tier participants systematically *under-react to the second derivative of news*: not to the headline itself, but to (a) how surprising it is relative to consensus, (b) how it interacts with current positioning (open interest, dealer gamma exposure), and (c) how it propagates across correlated assets before the Indian cash market fully reacts. MarketSage AI monetises that 30-second-to-30-minute interpretation lag — *not* the 30-millisecond wire race.

1.4 Success ladder

Tier	Name	Definition of done
T0	Instrument	System observes market and news, stores everything point-in-time. No prediction at all.
T1	Explainer	System explains *past* moves: "Nifty fell 0.8% at 11:04, attributed cause X" with more than 60% agreement from a human panel.
T2	Calibrated forecaster	Predicted probabilities match realised frequencies (reliability slope 0.9–1.1), even if the edge is thin.
T3	Positive expectancy	Paper trading over at least 3 months and at least 300 trades, positive after *full* costs, profit factor above 1.15.
T4	Live micro-capital	Real money, capped, ladder-scaled by live Sharpe.
T5	Product	Multi-user, latency SLO, compliance cleared.

Non-negotiable rule

You may not skip a tier. Most retail quant projects die because they jump from T0 straight to T4. Every phase gate in Chapter 17 exists to enforce this ladder.

CHAPTER 2

Problem Statement and Market Context

2.1 Who actually loses money and why

The Indian derivatives retail segment is the largest options market in the world by contract count, and the overwhelming majority of individual traders in index options lose money over a financial year. The loss mechanism is not usually a lack of information — it is four structural handicaps:

1. **Latency of interpretation.** Information reaches everyone almost simultaneously. What differs is the time to convert a headline into a position sizing decision. A human needs 4–10 minutes; by then, the first-order move is complete and the trader is buying the top.
2. **Volatility mispricing.** Retail buys options at elevated implied volatility right after news, then loses to volatility mean reversion even when direction is correct.
3. **No conditional base rates.** A trader has no idea that, historically, a Middle-East escalation headline of severity 7 produced a median Nifty move of -0.31% with an interquartile range of $[-0.9\%, +0.2\%]$ over 30 minutes — so they size as if the move were certain.
4. **No feedback loop.** Losses are not attributed. Nobody knows which event class they are systematically bad at.

MarketSage AI is explicitly designed as an answer to those four, in that order.

2.2 Where the edge can and cannot be

Edge type	Feasible for us?	Comment
Raw latency arbitrage (sub-ms)	**No**	Requires exchange co-location, FPGA, direct feeds. Capital and licensing far outside scope.
Order-book queue position	**No**	Same reason.
First-mention news detection (0.2–5 s)	**Partially**	Achievable for social and open-web sources. Premium wire feeds are priced for institutions.
Event interpretation and conditional base rates	**Yes**	This is our primary edge. It needs research, not co-location.
Cross-asset lead-lag (GIFT Nifty, DXY, Brent, US futures overnight)	**Yes**	Well documented, exploitable at minute granularity.
Volatility surface mispricing around events	**Yes**	Requires the option-chain data spine; strong, durable edge.
Positioning and dealer-gamma effects	**Yes**	Open interest is public; few retail systems model it properly.

Strategic conclusion

We are not building a high-frequency trading firm. We are building a **fast, disciplined, conditional-probability machine** that reacts in seconds where humans react in minutes, and that never takes a position whose expected value after costs is negative.

2.3 Scope decisions

Decision	Choice and rationale
Instruments (v1)	Nifty 50 and Bank Nifty index options only. Highest liquidity, tightest spreads, no single-stock idiosyncratic risk, no delivery obligation.
Horizons	1, 5, 15 and 30 minutes. Below 1 minute is microstructure noise we cannot win; above 60 minutes the event signal decays into macro drift.
Trade style	Intraday only in v1. Every position flat before close. Eliminates overnight gap risk, which is exactly the risk our models cannot forecast.
Direction of expansion	v2 adds sector baskets (Bank, IT, Energy, Pharma) via event-to-sector mapping; v3 adds overnight and gap-open strategies.
Execution	Human-in-the-loop signal until Tier T3 is passed. Auto-execution gated behind an explicit written checklist.

CHAPTER 3

Reality Calibration: What Is Actually Achievable

This chapter is deliberately pessimistic. Read it before writing a single line of code. If the project survives this chapter, it is worth building.

3.1 Latency: the honest budget

The claimed one second from tweet to trade is achievable, but the meaningful number is not our internal latency — it is our latency *relative to who has already traded*. Below is the realistic decomposition.

Stage	Target p50	Target p99	Notes
Source to our ingest (network)	250 ms	3000 ms	Uncontrollable. Social APIs poll or stream with variable delay; open-web scraping is worse.
Deduplication and normalisation	3 ms	15 ms	In-memory hash and embedding ANN lookup.
Tier-1 filter (rules + small model)	5 ms	25 ms	Rejects roughly 97% of traffic.
Tier-2 classification (transformer)	40 ms	120 ms	Batched GPU or quantised ONNX on CPU.
Impact scoring (lookup + Bayesian posterior)	4 ms	20 ms	Precomputed priors in Redis.
Feature assembly (online store)	8 ms	40 ms	Redis pipeline, pre-warmed.
Prediction ensemble	12 ms	60 ms	ONNX Runtime, batch size 1, warm.
Options engine (surface + Monte Carlo)	25 ms	90 ms	Vectorised, 20k paths.
Risk checks and publish	3 ms	12 ms	
Internal total (ingest to signal)	**100 ms**	**382 ms**	This is the number we control and must defend.
Wall-clock (event to signal)	**350 ms**	**3.4 s**	Dominated by source latency.

The uncomfortable truth about latency

Institutional participants using premium low-latency wire feeds will have acted before our first packet arrives. Therefore **our strategy must be profitable on the *residual* move, not the initial spike.** Every backtest must apply an artificial 3-second entry delay from event timestamp. If the edge disappears under that delay, the edge was never real.

3.2 Accuracy: what the literature and the data actually support

Task	Realistic OOS range	Interpretation
5-min direction, unconditional	50–52%	Essentially a coin flip. Do not trade this.
5-min direction, conditioned on a high-severity event	55–62%	The event conditioning is where the edge lives. Sample size is small, so confidence intervals are wide.
30-min direction after scheduled macro release	56–64%	Best regime: the surprise variable is measurable.
Realised volatility direction (up/down) over 15 min	62–70%	Volatility is far more forecastable than direction. **Exploit this.**
Sign of implied-volatility change post-event	65–75%	Even more forecastable. Strong basis for volatility structures.

Design implication that changes the whole product

Volatility is much more predictable than direction. Therefore the v1 money-maker should be **volatility structures** (long straddle before a scheduled event, short premium after a volatility spike, debit spreads when the forecast move is smaller than the implied move) with directional bets as the *secondary* strategy, not the primary one. The original design had this backwards.

3.3 The implied move: the single most important formula in this project

Before recommending any option trade, compute what the market has already priced in. For an at-the-money straddle with T years to expiry:

$$\text{Implied Move}_{\%} \approx \frac{C_{\text{ATM}} + P_{\text{ATM}}}{S} \qquad \text{or, from implied volatility,} \qquad \sigma_{\text{IV}}\sqrt{T}\,.$$

For an intraday horizon of h minutes with 375 trading minutes per session:

$$\text{Expected move over } h \;=\; S \cdot \sigma_{\text{IV}} \cdot \sqrt{\frac{h}{375 \times 252}}\,.$$

The trade filter

Let $\hat{\mu}_h$ be our forecast absolute move over horizon h and m_h the implied move over the same horizon. Then:

- $\hat{\mu}_h > 1.3\, m_h$ and directional conviction high $\Rightarrow$ **buy** a directional structure.
- $\hat{\mu}_h > 1.3\, m_h$ and direction uncertain $\Rightarrow$ **buy** volatility (straddle/strangle).
- $\hat{\mu}_h < 0.7\, m_h \Rightarrow$ **sell** volatility (defined-risk spreads only), or do nothing.
- Otherwise $\Rightarrow$ **no trade**. This will be the answer roughly 80% of the time, and that is correct behaviour.

3.4 The IV crush trap, illustrated

A trader sees a war headline, buys a 19500 PE at Rs. 240 with implied volatility at 22%. Nifty falls 0.6% in the next fifteen minutes — direction correct. But volatility mean-reverts to 16% as the market decides the event is contained.

Component	Delta effect	Vega effect	Theta
Spot moves -0.6% (≈ -117 pts), $\Delta = -0.45$	$+52.6$	—	—
IV falls $22\% \to 16\%$, vega ≈ 11 per volatility point	—	-66.0	—
15 minutes decay on expiry-week option	—	—	-4.5
Net	**-17.9** — *a loss on a correct call*		

Rule

No recommendation may be emitted without an explicit vega scenario. The engine must forecast $\Delta\sigma_{\text{IV}}$, not just ΔS. Chapter 11 specifies how.

3.5 Data reality

Data	Reality	Indicative cost	Decision
Live index/option ticks	Retail broker APIs are throttled (typically 1 snapshot per second, limited instrument subscriptions). True tick-by-tick needs a licensed vendor feed.	Broker API: free to low. Vendor tick feed: moderate monthly.	Start with broker API; upgrade at Phase 7.
Historical option chain (1-min, multi-year)	Not freely available. Must be purchased. This is the hard dependency for any credible options backtest.	Significant one-time or annual	**Buy at Phase 5.** Non-negotiable.
Point-in-time news with publish timestamps	Free APIs give limited history and unreliable timestamps.	Moderate	Build our own archive from day one; every day not collecting is a day of research data lost forever.
Social/short-form feeds	Paid API tiers; rate-limited	Moderate	Start with a curated account list, not the firehose.
Economic calendar with consensus	Essential for surprise features	Low	Required from Phase 2.
FII/DII, participant-wise OI	Published end of day by the exchange	Free	Daily batch ingest.

Start collecting today

The single highest-return action available right now is to start writing every tick, every option chain snapshot and every headline (with its publish timestamp) to disk. Models can be built later; **history cannot be recovered later.** Phase 0 exists for exactly this reason.

3.6 Regulatory reality (India)

- **Algorithmic trading via broker APIs** is governed by the SEBI framework for retail algorithmic trading. In practice this means: algorithms routed through a broker must be registered/tagged by that broker, orders carry a unique algo identifier, and static IP registration may be required. Confirm the current requirements directly with your chosen broker before any automated order is sent.
- **Trading your own capital** with your own tool is fundamentally different from **distributing recommendations to others**. The moment you give buy/sell recommendations to third parties, investment-adviser or research-analyst registration questions arise. The multi-user product in Tier T5 requires legal advice, not an engineering decision.
- **Exchange data redistribution** is licensed. Displaying live NSE prices to other users requires a data licence. Personal use is different from public display.
- **Source terms of service** for news and social APIs restrict storage and redistribution. Store identifiers plus derived features rather than raw content where terms require it.

Action item, not a footnote

Phase 0 includes a written compliance memo: chosen broker, their algo registration process, what you may store, and what you may show to others. Do not treat this as paperwork to be done later — it constrains architecture.

3.7 Design principles that follow

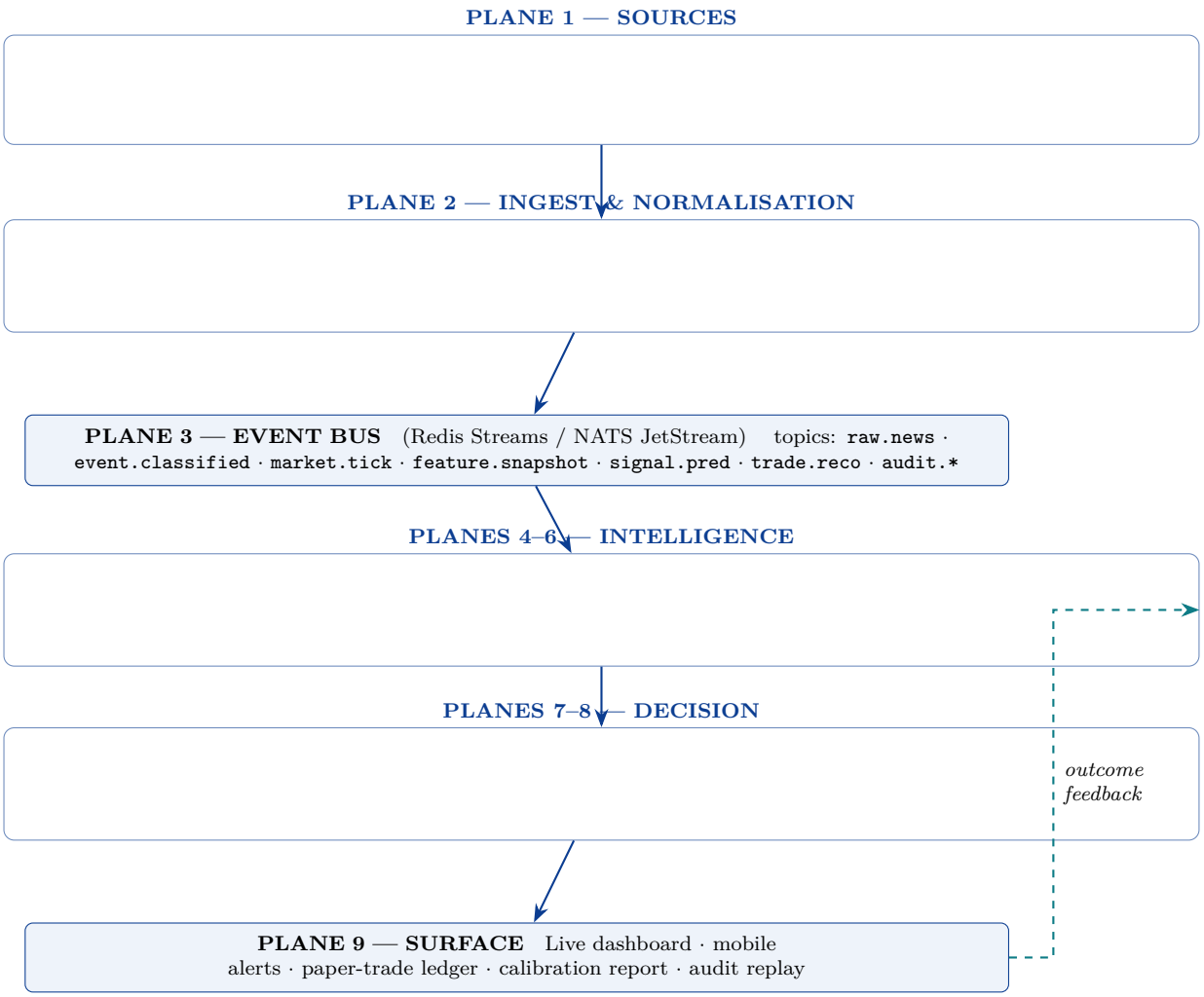
1. **Distributions, never point estimates.** Every model output is a distribution or a set of quantiles.
2. **Calibration outranks accuracy.** A 55% model that knows it is 55% is worth more than a 65% model that claims 90%.
3. **Cost-aware by construction.** Brokerage, STT, exchange fee, GST, stamp duty and slippage enter the objective function, not a post-hoc subtraction.
4. **No-trade is the default action.** The system's most common output must be silence.
5. **Point-in-time or it did not happen.** Any feature that could not have been computed at decision time is banned.
6. **Everything replayable.** Any past second of the market must be reconstructible bit-for-bit for debugging.
7. **Kill switches at every layer.** Automated systems fail loudly and expensively; they must fail closed.
8. **Simple baseline before deep model.** Gradient boosting on good features beats a poorly-fed neural network almost every time.

CHAPTER 4

System Architecture

4.1 The nine-plane view

The system is decomposed into nine planes. Each plane is independently deployable, independently testable, and communicates only through the event bus with versioned schemas. A plane may never reach into another plane's database.



4.2 Plane responsibilities and contracts

Plane	Owns	Hard contract it must honour
1 Sources	Nothing (external)	Every item must carry an original publish timestamp or be flagged `ts_uncertain`.
2 Ingest	Collectors, normaliser, dedup, candle builder	Emits only immutable, schema-versioned records. Never mutates history. Idempotent on replay.

Plane	Owns	Hard contract it must honour
3 Bus	Ordering, durability, replay	At-least-once delivery, consumer groups, 7-day retention hot, cold archive to Parquet.
4 Event	Taxonomy classification, entity extraction, novelty, credibility	Emits an `Event` with class, severity, confidence, entities, novelty and credibility. Never emits a trade opinion.
5 Impact	Historical conditional response distributions	Emits a distribution over index returns per horizon, plus sample size and confidence interval. Refuses to answer when $n < 8$.
6 Features	Online and offline feature computation	*Identical code path* for training and serving. Every feature carries an as-of timestamp.
7 Prediction	Direction, magnitude, volatility, quantiles	Outputs calibrated probabilities and conformal intervals. Must attach model version and feature hash.
8 Decision	Structure choice, sizing, risk	May veto anything. Owns all kill-switches. Writes an immutable decision record with full reasoning.
9 Surface	Presentation, alerting, ledger	Read-only with respect to trading state; may not influence decisions.

4.3 Two speed lanes

A recurring failure in such systems is trying to make everything fast. Instead, split explicitly:

	Hot lane (reflex)	Cold lane (deliberation)
Latency budget	Under 400 ms end to end	1–60 seconds
Triggered by	Breaking event, price shock, scheduled release	Every event, plus a 5-second heartbeat
Models used	Quantised, pre-warmed, small; rules and lookups	Full ensemble, LLM reasoning, cross-source confirmation, Monte Carlo
Output	Provisional alert with wide intervals	Full recommendation with structure and sizing
Can it place an order?	No. Never.	Yes (once Tier T4 is reached)
Rationale	Speed without confirmation is how you trade a fake headline	Confirmation without speed misses the move; two lanes get both

The two-lane rule

The hot lane says *"something is happening, widen your risk"*. The cold lane says *"here is the trade"*. Conflating them is the design error that turns a research system into an account-destroying machine.

4.4 Technology choices with justification

Layer	Choice	Why this, not the alternative
Language (services)	Python 3.12 + `asyncio` + `uvloop`	Ecosystem for ML is decisive. `uvloop` closes most of the gap for I/O bound work. Rewrite only the proven hot path later, if profiling demands it.
Language (hot path, later)	Rust (via PyO3)	Only for the tick handler and candle builder if p99 exceeds budget. Do not start here.
API	FastAPI	Async, typed, automatic OpenAPI, WebSocket support.
Event bus	Redis Streams (v1) → NATS JetStream (v2)	Redis Streams gives consumer groups and replay with zero extra infrastructure since Redis is already needed for the online feature store. Kafka is operationally heavy for a single-node start; NATS is the low-latency upgrade path.
Time-series store	PostgreSQL + TimescaleDB	Hypertables, continuous aggregates for candle roll-ups, compression, and plain SQL. Avoids a separate query language.
Research store	Parquet on local NVMe + DuckDB	Backtests are analytical scans, not point lookups. DuckDB over Parquet is dramatically faster and simpler than querying the OLTP database.
Online feature store	Redis (hashes + sorted sets)	Sub-millisecond reads, TTL semantics match feature freshness.
Vector search	FAISS or `pgvector`	Deduplication and nearest-historical-event retrieval. `pgvector` first for operational simplicity.
Model training	PyTorch, LightGBM, scikit-learn	LightGBM is the baseline that usually wins on tabular features; PyTorch for sequence and distribution models.
Model serving	ONNX Runtime	Removes Python overhead at inference; consistent across CPU and GPU; batch-1 latency is what matters here.
NLP models	A financial-domain encoder for sentiment/classification, a multilingual encoder for global sources, plus an LLM in the cold lane only	Encoders are fast and cheap; the LLM is reserved for reasoning where its latency is affordable.
Orchestration	Prefect or Dagster	Nightly retrains, data quality checks, backfills. Airflow is heavier than needed.
Experiment tracking	MLflow	Model registry plus metric lineage; the registry is what production reads from.
Frontend	React + TypeScript + a lightweight charting library	TradingView's free charting library is acceptable for candles; the fan-chart overlay is custom canvas work.
Transport to UI	WebSocket with binary framing (MessagePack)	JSON at 1 Hz per instrument is fine; at tick rate it is not.

Containerisation	Docker Compose (v1) → Kubernetes (only at multi-user)	Kubernetes before Tier T5 is pure overhead.
Monitoring	Prometheus + Grafana + structured JSON logs + OpenTelemetry traces	A trace ID must flow from the news item all the way to the recommendation.

Deployment location matters more than framework choice

Co-locate compute in the same cloud region as your broker/data endpoints (for NSE-facing infrastructure this generally means a Mumbai region). A poor region choice adds more latency than every micro-optimisation in this document combined.

4.5 Failure domains and degradation

Failure	Detection	Degraded behaviour
News source down	No heartbeat for 90 s	Mark source stale; reduce credibility weight of single-source events; continue on market-only features.
Market feed gap	Sequence gap or 3 s silence during market hours	Halt all new signals immediately. Flatten open paper positions at last known price. Alert.
Model server unhealthy	p99 latency or error rate breach	Fall back to the rules-only baseline model, flag every output as `DEGRADED`, forbid auto-execution.
Clock drift > 50 ms	NTP monitor	Refuse to emit signals. Time integrity is non-negotiable for point-in-time correctness.
Redis unavailable	Connection error	Hot lane stops; cold lane continues from TimescaleDB with a wider latency SLO.
Contradictory events	Two opposing high-severity events within 60 s	Emit `CONFLICT`; suppress all directional signals; volatility signals only.

CHAPTER 5

Data Layer

5.1 Source catalogue

Sources are tiered by credibility. Credibility is a numeric weight used later in impact scoring, not a binary trust decision.

Source class	Credibility c	Typical lag	Role
Central bank / regulator official release	1.00	0–2 s	Ground truth for policy events. Scrape the official page directly, not a news summary of it.
Exchange notices	1.00	0–5 s	Halts, bans, margin changes.
Major international wires	0.90	1–10 s	Primary breaking coverage.
Verified official accounts of heads of state / ministries	0.85	0–3 s	Highest-impact, highest-noise. Requires source-specific handling.
Tier-1 national business media	0.75	30–180 s	Confirmation, colour.
Aggregator APIs	0.60	60–600 s	Breadth, not speed.
Anonymous social accounts	0.15–0.35	0–5 s	Early warning only. May *never* independently trigger a directional trade.
Forums and community feeds	0.10	Variable	Sentiment aggregate only.

5.1.1 Market data

Stream	Frequency	Use
Index spot (Nifty, Bank Nifty, Fin Nifty)	Tick or 1/s	Base price series, candle construction.
Index futures	Tick or 1/s	Basis, and it often leads spot.
Full option chain (both indices, near two expiries)	1–5 s snapshot	Implied volatility surface, open interest, put-call ratio, max pain, gamma exposure.
India VIX	1/s	Regime variable, volatility term structure anchor.
GIFT/SGX Nifty	Tick	Overnight and pre-open leading indicator.
Global futures (US equity index, crude, gold), DXY, USDINR	1/s or 1-min	Cross-asset propagation, overnight event pricing.
Sector indices	1/s	Event-to-sector routing.
Participant-wise open interest, FII/DII flows	End of day	Positioning regime features.

Corporate actions, expiry calendar, holiday calendar	Static/daily	Absolutely required to avoid silent backtest corruption.

5.2 The candle builder (and why it is subtle)

Building 1-minute candles from ticks looks trivial and is the most common source of quiet data corruption. Requirements:

1. **Exchange time, not local time.** Every timestamp normalised to exchange time with explicit timezone; store as UTC with an exchange-time view.
2. **Left-closed, right-open intervals.** A tick at exactly 09:20:00.000 belongs to the 09:20 candle, not 09:19.
3. **Explicit gap handling.** If no tick arrives in a minute, emit a zero-volume candle with $O = H = L = C =$ previous close, flagged `synthetic`. Silent omission destroys sequence models.
4. **Session boundaries.** Pre-open (09:00–09:15) is a different price formation mechanism; tag it, never mix it into continuous-session statistics.
5. **Late ticks.** A tick arriving after its candle is sealed goes to a correction log; the candle is versioned, not overwritten in place.
6. **Determinism.** Replaying the same tick file must yield a byte-identical candle file. Enforce this with a test.

5.3 Point-in-time discipline

The three timestamps

Every record carries three timestamps, always:

- `t_event` — when the thing happened in the world (publish time).
- `t_ingest` — when our system first saw it.
- `t_available` — when it became usable for a decision (after dedup, classification and feature computation).

Backtests may only use data where `t_available` $\leq$ the decision time. A feature computed from `t_event` alone is look-ahead bias wearing a disguise.

Revised data (an economic figure corrected an hour later, an option chain snapshot re-published) is stored as a new row with a validity interval, never as an update. The research layer always reads "the value as known at time t".

5.4 Storage layout

Listing 5.1: Core TimescaleDB schema (abridged)

```
-- Ticks: hypertable partitioned by day, compressed after 7 days
CREATE TABLE tick (
  ts TIMESTAMPTZ NOT NULL,
  instrument_id INT NOT NULL,
  ltp NUMERIC(12,2) NOT NULL,
  qty INT,
  bid NUMERIC(12,2), ask NUMERIC(12,2),
  bid_qty INT, ask_qty INT,
  oi BIGINT,
```

```
  seq BIGINT NOT NULL, -- gap detection
  src SMALLINT NOT NULL
);
SELECT create_hypertable('tick','ts', chunk_time_interval => INTERVAL '1 day');

-- Candles: one row per (instrument, timeframe, bucket)
CREATE TABLE candle (
  ts TIMESTAMPTZ NOT NULL, instrument_id INT NOT NULL, tf SMALLINT NOT NULL,
  o NUMERIC(12,2), h NUMERIC(12,2), l NUMERIC(12,2), c NUMERIC(12,2),
  v BIGINT, oi BIGINT, trades INT,
  vwap NUMERIC(12,2), synthetic BOOLEAN DEFAULT FALSE, version SMALLINT DEFAULT 1,
  PRIMARY KEY (instrument_id, tf, ts, version)
);

-- Option chain snapshots
CREATE TABLE option_snap (
  ts TIMESTAMPTZ NOT NULL, underlying_id INT, expiry DATE,
  strike NUMERIC(10,2), opt_type CHAR(2),
  ltp NUMERIC(10,2), bid NUMERIC(10,2), ask NUMERIC(10,2),
  volume BIGINT, oi BIGINT, oi_change BIGINT,
  iv NUMERIC(6,3), delta NUMERIC(6,4), gamma NUMERIC(10,8),
  theta NUMERIC(8,4), vega NUMERIC(8,4), rho NUMERIC(8,4),
  spot NUMERIC(12,2)
);

-- News / event store: append-only, point-in-time
CREATE TABLE news_raw (
  id UUID PRIMARY KEY,
  t_event TIMESTAMPTZ, t_ingest TIMESTAMPTZ NOT NULL, ts_uncertain BOOLEAN,
  source_id INT NOT NULL, url TEXT, lang CHAR(3),
  headline TEXT, body TEXT, content_hash BYTEA, simhash BIGINT,
  embedding vector(768)
);

CREATE TABLE event_classified (
  id UUID PRIMARY KEY, news_id UUID REFERENCES news_raw(id),
  t_available TIMESTAMPTZ NOT NULL,
  event_class TEXT NOT NULL, subclass TEXT,
  severity NUMERIC(3,1), confidence NUMERIC(4,3),
  novelty NUMERIC(4,3), credibility NUMERIC(4,3),
  entities JSONB, geo JSONB, surprise NUMERIC(8,4),
  model_version TEXT NOT NULL
);
```

Data	Hot store	Cold store / research
Ticks	TimescaleDB 7 days	Daily Parquet, partitioned `/date=/instrument=`
Candles	TimescaleDB, all	Parquet mirror for DuckDB scans
Option chain	TimescaleDB 30 days	Parquet, partitioned by date and expiry
News & events	PostgreSQL, all	Parquet + embeddings index
Features	Redis (TTL 15 min)	Parquet feature snapshots per decision
Decisions & outcomes	PostgreSQL, all	Never deleted. This is the audit spine.

5.5 Data quality gates (run continuously)

Check	Threshold	Action on breach
Tick sequence gaps	any gap	Log, backfill from broker history, mark affected minute untrusted.

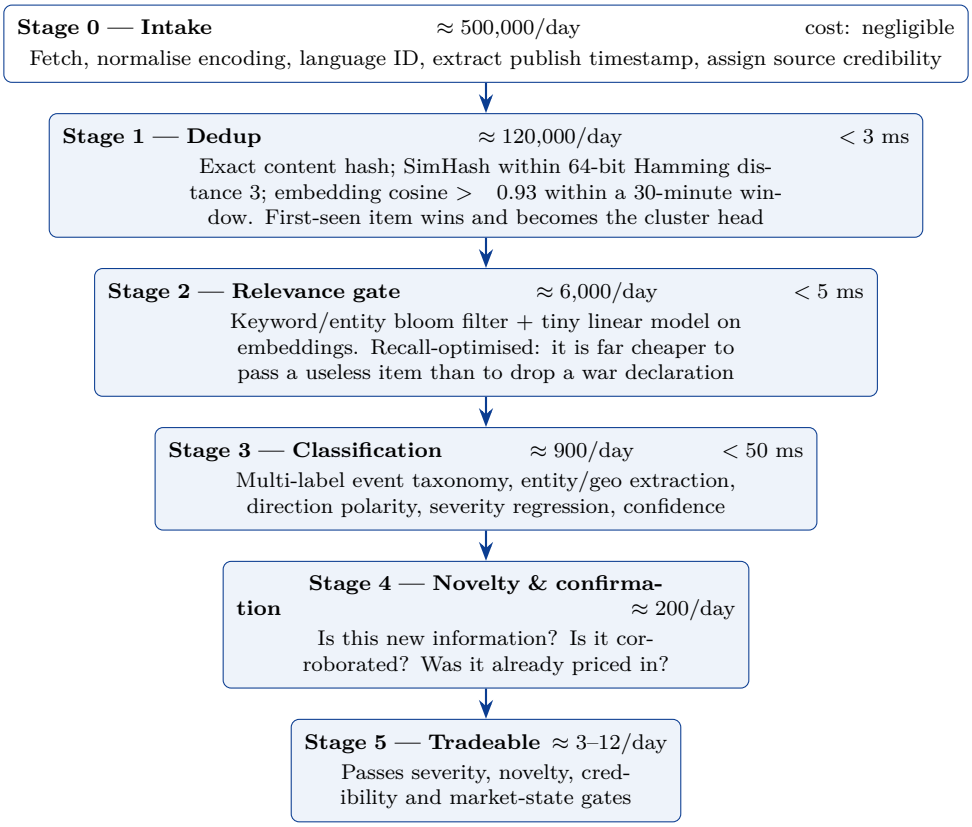
Candle count per session	exactly 375 for continuous session	Block research jobs on that day until resolved.
Option chain completeness	$\geq$ 95% of expected strikes	Do not compute surface; mark surface stale.
Implied volatility sanity	$0.03 < \sigma < 3.0$, no-arbitrage monotonic in strike	Reject the quote, use surface fit instead.
Put-call parity deviation	beyond spread-implied bound	Flag stale quote; do not trade that strike.
News timestamp sanity	`t_event` $\leq$ `t_ingest` and within 7 days	Set `ts_uncertain`, exclude from latency-sensitive research.
Duplicate rate	$< 3\%$ post-dedup	Retune dedup thresholds.
Feature null rate	$< 1\%$ per feature per day	Block model training.
Train/serve skew	feature distribution PSI < 0.2	Alert; investigate before next retrain.

CHAPTER 6

Event Detection Engine

6.1 The funnel

Roughly 500,000 items per day may enter the pipeline. Perhaps 40 are worth a model's attention and 3 are worth a trade. The engine is therefore a funnel with increasing cost per stage.



Asymmetric error policy

Stages 0–2 are tuned for **recall** (never lose a real event). Stages 3–5 are tuned for **precision** (never trade a non-event). Mixing these objectives in one model is the classic mistake.

6.2 Event taxonomy

The taxonomy is the backbone of the entire system: impact priors, features, model conditioning and post-trade attribution are all keyed on it. It must be designed once, carefully, and versioned.

Family	Classes (examples)	Typical severity	Primary Indian transmission channel
Monetary policy	rate decision, stance change, liquidity measure, unscheduled action, minutes/commentary	6–10	Bank Nifty directly; rate-sensitive sectors; currency

Family	Classes (examples)	Typical severity	Primary Indian transmission channel
Macro release	inflation, growth, employment, trade balance, industrial output, fiscal data	4–8	Broad index via surprise vs consensus
Geopolitical	military action, escalation rhetoric, territorial incident, sanctions, treaty, coup	3–10	Risk premium, crude, currency, gold; defence and oil-marketing sectors
Trade & tariffs	tariff announcement, export ban, supply-chain restriction	4–9	Export sectors (IT, pharma, chemicals, textiles)
Energy	production quota change, supply disruption, strategic reserve action, pipeline/refinery incident	4–9	Crude to inflation to rates; paint, aviation, oil marketing, tyres
Domestic policy	budget item, regulatory change, sector reform, tax change, subsidy	3–9	Sector-specific, occasionally index-wide
Corporate/market structure	index rebalancing, large block deal, big listing, margin/regulatory change	3–7	Flow-driven, highly mechanical, often the most predictable class
Flows & positioning	foreign flow reversal, large open-interest shift, expiry-day pinning	3–6	Mechanical, intraday, exploitable
Disaster & health	natural disaster, epidemic, major infrastructure failure	2–9	Insurance, infrastructure, supply chains
Political	election result, coalition change, major resignation, scandal	3–10	Broad risk premium; extremely high variance
Noise / non-event	opinion, analysis, repetition, speculation	0–2	**Explicit class** — the model must learn to say "nothing here"

Each classified event carries the following fields.

Field	Type	Meaning and how it is produced
`class`, `subclass`	enum	Multi-label transformer classifier.
`polarity`	$[-1, 1]$	Risk-off (-1) to risk-on ($+1$) for Indian equities specifically — *not* generic sentiment. A rate cut is negative sentiment about growth but positive polarity for equities.
`severity`	$[0, 10]$	Regression head trained on human-labelled anchors; described in Section 6.3.
`confidence`	$[0, 1]$	Classifier probability, temperature-calibrated.
`entities`	list	Countries, institutions, commodities, companies, sectors — linked to a canonical knowledge graph, not free text.

`geo`	list	Location plus a distance/exposure weight to India.
`novelty`	$[0, 1]$	Section 6.4.
`credibility`	$[0, 1]$	Source weight, boosted by corroboration.
`surprise`	$\mathbb{R}$	For scheduled events: $(\text{actual} - \text{consensus})/\sigma_{\text{consensus}}$. Null otherwise.
`expected`	bool	Was this event on the calendar? Scheduled and unscheduled events behave completely differently.

6.3 Severity: how to make it real and not arbitrary

A hand-written severity table will not survive contact with reality. Build severity in three passes:

1. **Anchor set.** Hand-label 400–600 historical headlines on a 0–10 scale using written rubrics with concrete anchors (10 = existential shock; 7 = major unexpected policy shift; 4 = meaningful but priced; 1 = commentary). Two labellers, measure inter-annotator agreement, resolve disagreements.
2. **Supervised head.** Train a regression head on the encoder to predict the anchor scores.
3. **Empirical recalibration.** This is the step everyone skips: regress the predicted severity against the *realised* absolute abnormal move of Nifty in the following 30 minutes across all historical events. Then remap severity so that it is monotonic in realised impact. Severity becomes an *empirically grounded* quantity rather than an opinion.

$$\text{severity}^{\text{cal}} = F^{-1}_{|r_{30}|}\Big(\hat{F}_{\text{sev}}\big(\text{severity}^{\text{raw}}\big)\Big)$$

where $\hat{F}_{\text{sev}}$ is the empirical CDF of raw severity and $F_{|r_{30}|}$ the empirical CDF of absolute 30-minute abnormal returns.

6.4 Novelty: is this actually new information?

The most expensive mistake is trading a re-report of something already priced. Novelty combines four signals:

$$\begin{aligned}\text{novelty} \;=\; & w_1 \underbrace{\big(1 - \max_j \cos(e, e_j)\big)}_{\text{semantic distance to last 6 h}} + w_2 \underbrace{\mathbb{1}[\text{first mention in cluster}]}_{\text{primacy}} \\ & + w_3 \underbrace{\big(1 - \rho_{\text{cal}}\big)}_{\text{not on calendar}} + w_4 \underbrace{\big(1 - |\hat{r}_{\text{pre}}|/\sigma_5\big)}_{\text{market has not already moved}}\end{aligned}$$

The fourth term is the most underrated: if the index has already moved two standard deviations in the five minutes *before* our detection, the information is in the price and the opportunity has passed. In that case the system should consider a *fade* strategy or, more usually, no trade at all.

6.5 Confirmation and the fake-news defence

Rule	**Condition**	**Consequence**
Single low-credibility source	$c < 0.5$, no corroboration	Hot-lane alert only. No directional signal. Volatility-widening only.

Corroboration window	Second independent source with $c \geq 0.75$ within 120 s	Credibility upgraded; directional signal permitted.
Official confirmation	Primary source (regulator, central bank, exchange, official channel)	Full weight immediately.
Contradiction	An equally credible source contradicts within 5 min	Emit `RETRACTION`; immediately close any position opened on that event and mark the event poisoned.
Extreme severity gate	severity ≥ 9	Requires **two** independent sources with $c \geq 0.8$ regardless of speed. The most extreme headlines are the most likely to be fake.
Market disagreement	Signal says down, but index, futures and currency all move up within 30 s	Suppress. The market is a better classifier than we are.

The AP-hack test

Before going live, run this test: inject a synthetic, highly credible-looking but false catastrophic headline from a single medium-credibility source into a replay. **The system must not produce a directional trade.** If it does, the confirmation logic is wrong. Repeat this test in CI forever.

6.6 Entity-to-instrument routing

An event mentioning "crude supply disruption" must mechanically become an exposure vector. Maintain a knowledge graph:

Listing 6.1: Exposure routing (conceptual)

```
# knowledge graph edge: (entity) --affects--> (factor) --loads_on--> (instrument)
EXPOSURE = {
    "crude_oil": {"NIFTY": -0.25, "BANKNIFTY": -0.15,
                  "sector:OIL_MKTG": -0.80, "sector:AVIATION": -0.70,
                  "sector:PAINTS": -0.55, "sector:UPSTREAM": +0.60},
    "usd_inr_up": {"NIFTY": -0.20, "sector:IT": +0.65, "sector:PHARMA": +0.40},
    "rbi_rate_cut": {"BANKNIFTY": +0.70, "NIFTY": +0.45,
                     "sector:REALTY": +0.75, "sector:AUTO": +0.55},
}
# Beta values are ESTIMATED FROM HISTORY (rolling regression), never hand-typed
# in production. The dict above is only the bootstrap prior for cold start.
```

Rule

Hand-written coefficients are acceptable only as a cold-start prior. Every coefficient must be replaced by a rolling empirical estimate with a confidence interval as soon as sufficient history exists. The system must be able to answer "how do you know?" with a number and a sample size.

CHAPTER 7

Impact Scoring Engine

7.1 What impact scoring actually is

The original design computes "Nifty will drop 1.5%". That is a point estimate with no error bar, and it is the root cause of over-sizing. The correct object is a **conditional predictive distribution**:

$$P(r_{t\to t+h} \mid \text{event class } c,\ \text{severity } s,\ \text{surprise } z,\ \text{market state } m)$$

produced primarily from historical evidence, and used as a *prior* that the ML models in Chapter 9 then update with live microstructure.

7.2 Building the impact library: event study methodology

For every historical event of class c:

1. Define the event time t_0 precisely (publish timestamp, exchange time).
2. Compute the **abnormal return** over horizon h: subtract the expected return given the market's normal behaviour at that time of day and volatility regime,

$$AR_h = r_{t_0\to t_0+h} - \hat{\mathbb{E}}[r_h \mid \text{time-of-day}, \sigma_{\text{regime}}, \text{day-type}]\,.$$

   Time-of-day normalisation matters enormously in Indian intraday data: the volatility profile is strongly U-shaped, so a raw 0.3% move at 09:20 is ordinary while the same move at 13:30 is extreme.
3. Compute a matched **control distribution**: sample non-event windows with the same time-of-day and volatility regime.
4. Record: AR_h for $h \in \{1, 5, 15, 30, 60\}$ minutes, realised volatility ratio, implied volatility change, maximum favourable and adverse excursion, volume ratio, and time-to-peak.
5. Test significance against the control distribution with a bootstrap, not a normal approximation (intraday returns are fat-tailed).

Stored statistic	**Why it matters for trading**
Median AR_h and quantiles	The realistic expectation and its spread.
P(sign correct)	Whether a directional trade is even sensible.
Realised-vol ratio $\sigma_{\text{post}}/\sigma_{\text{pre}}$	Whether to buy volatility.
$\Delta\sigma_{\text{IV}}$ distribution	**The vega decision.** Determines whether options get more or less expensive.
Max favourable / adverse excursion	Where to put the target and the stop, from evidence rather than a round number.
Time-to-peak	The correct holding period; also determines the horizon to trade.
Half-life of the effect	Whether the move persists or mean-reverts.
Sample size n	The honesty check.

7.3 Hierarchical Bayesian shrinkage for rare events

Most interesting events are rare. With $n = 4$ historical instances, the empirical mean is noise. Use a hierarchical model that shrinks class-level estimates toward their family-level estimate:

$$\mu_c \sim \mathcal{N}(\mu_{\text{family}}, \tau^2), \qquad AR_{c,i} \sim \mathcal{N}(\mu_c \cdot g(s_i), \sigma_c^2)$$

with posterior mean approximately

$$\hat{\mu}_c = \frac{n_c/\sigma_c^2}{n_c/\sigma_c^2 + 1/\tau^2}\,\bar{AR}_c \;+\; \frac{1/\tau^2}{n_c/\sigma_c^2 + 1/\tau^2}\,\mu_{\text{family}}.$$

Practical consequence

With $n_c = 3$, the estimate is dominated by the family prior — which is exactly correct behaviour. The engine reports the posterior *and* the sample size, and the risk gate reduces position size when the posterior variance is high. Refuse to emit a class-specific impact when $n_c < 8$ and no family prior exists; fall back to "unknown event, volatility-only response".

7.4 The live scoring formula

$$\hat{r}_h = \underbrace{\hat{\mu}_{c,h}}_{\text{Bayesian prior}} \times \underbrace{g(s)}_{\text{severity}} \times \underbrace{\nu}_{\text{novelty}} \times \underbrace{c_{\text{cred}}}_{\text{credibility}} \times \underbrace{\kappa(m)}_{\text{regime}} \times \underbrace{\lambda(\tau)}_{\text{time-of-day}} \times \underbrace{(1-\pi)}_{\text{already-priced discount}}$$

Term	Form	Notes
$g(s)$	$(s/5)^\gamma$, $\gamma \approx 1.4$ fitted	Impact is convex in severity: a 9 is much more than twice a 4.5.
ν	novelty score	Old news gets multiplied toward zero.
c_{cred}	credibility	Unconfirmed rumour is heavily discounted.
$\kappa(m)$	regime multiplier	Same event has larger impact in a high-volatility, low-liquidity regime. Fitted per regime cluster.
$\lambda(\tau)$	time-of-day multiplier	U-shaped; also handles the special cases of the opening 15 minutes, the pre-close and expiry-day afternoons.
π	$\min(1, \lvert r_{\text{pre}}\rvert / \lvert\hat{\mu}_{c,h}\rvert)$	If the market has already made the expected move before we act, the remaining opportunity is gone. **This term is what prevents chasing.**

The output is not a scalar but a triple: ($\hat{r}_h$, quantiles, n_{eff}) for each horizon h, plus a separate volatility forecast ($\hat{\sigma}_{\text{post}}/\sigma_{\text{pre}}$, $\widehat{\Delta\sigma_{IV}}$).

7.5 Cross-asset propagation

When an event occurs outside Indian market hours, or affects a foreign asset first, the impact reaches Nifty through a measurable chain. Estimate the chain empirically with lagged regressions and store the transfer coefficients:

Channel	Typical lag	Modelling note

GIFT/SGX Nifty to Nifty spot	Near-instant during overlap; sets the gap when Indian market is shut	The single most useful overnight variable.
US equity futures to Indian open	Overnight	Estimate a gap model: $r_{\text{gap}} = \alpha + \beta\, r_{\text{US futures}} + \ldots$
Crude to Nifty	Minutes to hours	Asymmetric: sharp spikes transmit faster than slow drifts.
USD/INR to Nifty	Minutes	Bidirectional causality; use a vector autoregression, not a simple regression.
US Treasury yields to Bank Nifty	Hours	Slower; more relevant to the next session.
Asian indices to Nifty intraday	Minutes	Useful during the 09:15–12:00 window.

The overnight opportunity

Roughly half of the total price change in Indian indices arrives as overnight gaps. Most retail systems ignore this because they only look at intraday candles. A well-calibrated **gap forecaster** driven by overnight global events is arguably an easier and more profitable first product than intraday prediction — and it needs no low-latency infrastructure at all. Consider shipping it first (see Phase 4 in the roadmap).

CHAPTER 8

Feature Store and Feature Catalogue

8.1 Why a feature store and not a script

The most common cause of a backtest that works and a live system that does not is **train/serve skew**: the offline feature was computed differently from the online one. The rule is absolute.

One definition, two materialisations

A feature is defined *once*, in one function, with an explicit dependency list and an as-of timestamp. The offline path runs it over Parquet history; the online path runs the same function over Redis state. A CI test replays a live day offline and asserts every feature matches to within floating-point tolerance. If it does not match, the build fails.

Listing 8.1: Feature definition pattern

```python
@feature(name="ofi_60s", version=3, deps=["book.bid","book.ask"],
        window="60s", ttl="5s", fill="ffill_max_2")
def order_flow_imbalance(book: BookWindow) -> float:
    """Signed order-flow imbalance over the trailing 60 seconds.
    Point-in-time safe: reads only data with ts <= as_of."""
    ...

# Offline: engine.materialise(ofi_60s, start, end) -> Parquet column
# Online : engine.serve(ofi_60s, as_of=now) -> float from Redis
# CI : assert_offline_online_parity(ofi_60s, sample_day)
```

8.2 Feature catalogue

8.2.1 Group A — Price and trend (the baseline)

Feature	Windows	Rationale
Log returns	1, 2, 5, 15, 30, 60 min	Raw momentum at several scales.
Return z-score vs time-of-day baseline	same	Normalises the U-shaped intraday volatility profile. **Critical.**
Realised volatility (Parkinson, Garman–Klass, Rogers–Satchell)	5, 15, 30, 60 min	Range-based estimators are far more efficient than close-to-close.
Bipower variation and jump component	30 min	Separates continuous volatility from jumps; events cause jumps.
Distance from VWAP, from opening range, from previous close	session	Standard intraday anchors that real traders actually watch.
Position within the day's range	session	Mean-reversion versus breakout context.
Trend strength / efficiency ratio	15, 30 min	Distinguishes a directional move from chop; regime information.
Autocorrelation of 1-min returns	30 min	Negative values signal a mean-reverting microstructure regime.

8.2.2 Group B — Microstructure (where the short-horizon edge lives)

Feature	Windows	Rationale
Order-flow imbalance (OFI)	10, 30, 60 s	Best-in-class short-horizon directional predictor when book data is available.
Bid–ask spread and its z-score	30 s	Widening spreads precede volatility and raise transaction cost — feeds both the model and the cost model.
Depth imbalance at top levels	instantaneous	Directional pressure.
Trade-sign imbalance (tick rule / Lee–Ready)	60 s	Aggressor direction.
Volume-clock and VPIN-style toxicity proxy	50 buckets	Flow toxicity spikes before large moves.
Large-trade ratio	60 s	Institutional participation proxy.
Quote update intensity	10 s	Attention and uncertainty proxy.
Futures basis (future minus spot)	instantaneous	Basis moves often lead spot.

Availability caveat

Full order-book depth may not be available on a retail broker feed. Design the model so Group B is optional: train two variants, with and without microstructure, and let the serving layer select by feed capability. Never train on features you will not have at 09:16 on a live Tuesday.

8.2.3 Group C — Options-derived (the most differentiated group)

Feature	Update	Rationale
ATM implied volatility, both expiries	1–5 s	Level of expected volatility.
IV term structure slope (near vs next expiry)	5 s	Inversion signals expected near-term event risk.
25-delta risk reversal (skew)	5 s	The market's directional fear; one of the strongest single option-derived predictors.
Butterfly / smile convexity	5 s	Tail pricing.
IV rank and IV percentile	daily	Is volatility cheap or expensive in context?
Put-call ratio, by volume and by open interest	1 min	Positioning and sentiment.
Open-interest change by strike, call and put	1 min	Where new positions are being built — writers define support and resistance.
Max pain and distance to it	5 min	Expiry-week pinning effect.
Estimated dealer gamma exposure (GEX)	1 min	Positive gamma dampens moves, negative gamma amplifies them. **Regime-defining.**

Implied move for the remaining session	1 min	The denominator of the trade filter in Chapter 3.
Volatility risk premium (IV minus forecast realised)	5 min	The core edge for volatility structures.

8.2.4 Group D — Cross-asset and global

Overnight and same-session values of: global equity futures, regional indices, crude, gold, dollar index, USD/INR, US yields, and a global-risk composite (first principal component of a basket of risk assets). Include both level changes and z-scores versus their own trailing distributions.

8.2.5 Group E — Event features

Feature	Note
Event class one-hot / embedding	Learned embedding is better once enough history exists.
Calibrated severity, polarity, novelty, credibility, confidence	From Chapter 6.
Surprise z-score	For scheduled releases; the single most powerful macro feature.
Prior impact posterior (mean, spread, n_{eff})	From Chapter 7. Feeding the prior in as a feature lets the model learn when to trust it.
Seconds since event	The decay clock.
Event density in trailing 30 min	Multiple simultaneous events change dynamics.
Conflict flag	Opposing events present.

8.2.6 Group F — Calendar, regime and session

Minutes since open, minutes to close, day of week, days to weekly expiry, days to monthly expiry, expiry-day flag, budget/policy-day flag, holiday-adjacent flag, current regime label from the regime model, trailing 20-day realised volatility percentile, and recent trend regime.

Feature hygiene

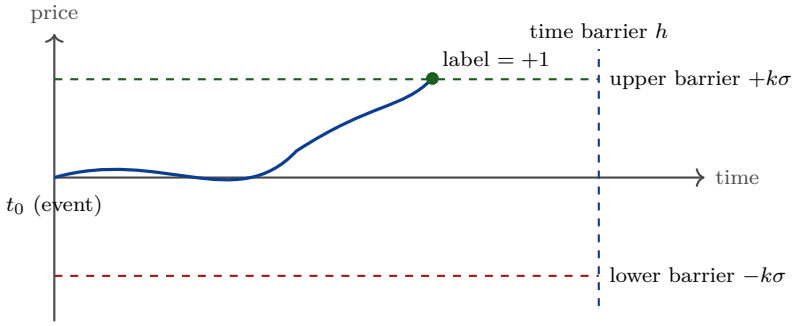
Target roughly 120–250 engineered features, not thousands. Run **Purged Group Time-Series CV** feature importance and drop anything that does not survive. High-cardinality raw features and features with any dependence on future information are removed automatically by a leakage linter that checks each feature's dependency timestamps.

CHAPTER 9

Prediction Models

9.1 Labelling: the decision that determines everything

Predicting "next candle up or down" is the wrong label because it does not correspond to a tradeable decision. Use the **triple-barrier** method.



Choice	Specification
Barriers	Upper and lower at $\pm k\sigma_t$ where σ_t is the *current* volatility estimate (not a fixed point value). k tuned per horizon; start with $k = 1.5$.
Time barrier	$h \in \{5, 15, 30\}$ minutes and always before the session close.
Label	$+1$ upper hit first, -1 lower hit first, 0 time barrier hit.
Sample weights	Weight each sample by uniqueness (overlapping outcome windows are not independent) and by absolute realised return.
Meta-labelling	A second model predicts *whether the primary model's call should be taken.* Primary model decides side; meta model decides size, including size zero. This consistently improves precision.

Why triple-barrier changes the product

It directly encodes stop-loss and target into the label. A model trained this way is answering the question you actually care about — "if I enter now with this stop and this target, do I win?" — rather than an academic question about candle colour.

9.2 Cross-validation without lying to yourself

Technique	Why it is mandatory
Purged K-fold	Remove training samples whose outcome window overlaps the test window. Without purging, the model sees the answer.
Embargo	Additionally drop a buffer (e.g. 1 trading day) after each test fold to kill serial-correlation leakage.
Walk-forward (anchored and rolling)	The only evaluation that resembles production. Report both.
Combinatorial purged CV	Generates many backtest paths, giving a *distribution* of Sharpe ratios rather than one lucky number.

Deflated Sharpe ratio	Corrects for the number of strategy variants tried. If you tested 200 configurations, your best Sharpe is inflated — quantify it.
Event-level splitting	Never split the same event's samples across train and test. Group by event ID.

Leakage checklist — run before believing any result

1. Does any feature use data with a timestamp after the decision time?
2. Is normalisation (mean/std, scaler) fitted on the full dataset instead of train only?
3. Are volatility estimates computed with a centred window?
4. Are labels from overlapping windows present in both train and test?
5. Was feature selection done before the split?
6. Does the news timestamp reflect publication or the time you downloaded it?
7. Is the option chain snapshot the one available at decision time, or the end-of-day file?
8. Are corporate actions and index reconstitutions handled point-in-time?

Any single "yes" invalidates the result. Most spectacular backtests die here.

9.3 Model portfolio

Do not build one model. Build a small ensemble of models with genuinely different inductive biases, then combine them.

Model	Predicts	Role and notes
M0 Rules baseline	direction	Simple event-prior lookup plus momentum filter. **Every ML model must beat this or it does not ship.** Also the degraded-mode fallback.
M1 LightGBM	$P(\text{up})$, triple-barrier class	The workhorse. Handles tabular heterogeneous features and missing values, trains in minutes, is interpretable via SHAP. Expect this to be your best model for a long time.
M2 Quantile LightGBM	$q_{05}, q_{25}, q_{50}, q_{75}, q_{95}$ of r_h	Pinball loss. Feeds the fan chart directly.
M3 Temporal CNN / TCN	direction and magnitude	Learns local shape patterns from raw candle and microstructure sequences that handcrafted features miss.
M4 Sequence transformer	multi-horizon distribution	Attention over the last N minutes plus event tokens. Highest ceiling, highest overfitting risk, needs the most data. **Build this last, not first.**
M5 Volatility model	σ_{t+h}	HAR-RV baseline plus a gradient-boosting correction on the residual. Volatility is the most forecastable target — give it a dedicated, well-tuned model.

M6 model	IV-change	$\Delta\sigma_{IV}$	Predicts implied volatility movement. **Directly determines options profitability.** Features: current IV rank, event class, term-structure slope, time to expiry, realised-implied spread.
M7	Meta-labeller	take / skip	Trained on primary model outputs plus market state. Its job is to say no.
M8	Regime classifier	regime label	Hidden Markov model or clustering over volatility, trend and liquidity. Everything else is conditioned on it.

9.4 Ensembling and calibration

1. **Stacking with time-aware folds.** Out-of-fold predictions from M1–M4 become inputs to a simple logistic regression meta-learner. Keep the meta-learner simple; a complex stacker overfits immediately.
2. **Regime-conditional weights.** Model weights vary by regime from M8; a transformer that shines in trending regimes may be useless in chop.
3. **Probability calibration.** Fit isotonic regression or Platt scaling on a held-out, chronologically later slice. Then verify with a reliability diagram and the Brier score decomposition (reliability, resolution, uncertainty).
4. **Conformal prediction.** Wrap the magnitude forecast in split conformal intervals to obtain finite-sample coverage guarantees. When the interval is too wide to be tradeable, the honest answer is no trade — and now the system can prove it.

The calibration contract

The dashboard's headline metric is not accuracy. It is the reliability diagram: of all the times the system said 60%, how often was it right? A system that is calibrated can be sized correctly. A system that is accurate but overconfident will still blow up.

9.5 Training loop and retraining policy

Job	Cadence	Detail
Nightly feature materialisation	Daily 18:00	Recompute the day's features, run quality gates, append to Parquet.
Incremental retrain	Weekly (weekend)	Refit M1, M2, M5, M6 on the expanded window; run walk-forward; promote only if it beats the incumbent on the last 4 weeks out-of-sample.
Full retrain and architecture search	Quarterly	Includes M3/M4 hyperparameters and feature re-selection.
Impact library rebuild	Weekly	New events added; posteriors updated.
Calibration refresh	Weekly	Isotonic refit on the most recent 4 weeks.
Drift monitor	Continuous	PSI on features, KL divergence on prediction distribution, rolling Brier score. Breach triggers alert and blocks auto-execution.

Champion/challenger	Continuous	The challenger runs in shadow mode on live data for at least 15 sessions before it may be promoted.

Anti-overfitting discipline

Keep a **locked holdout**: the most recent 3 months of data, encrypted and untouched. You may evaluate against it *once*, at the Phase 7 gate. Every additional peek burns the only honest estimate you will ever have. Log every experiment; deflate your Sharpe by the number of trials.

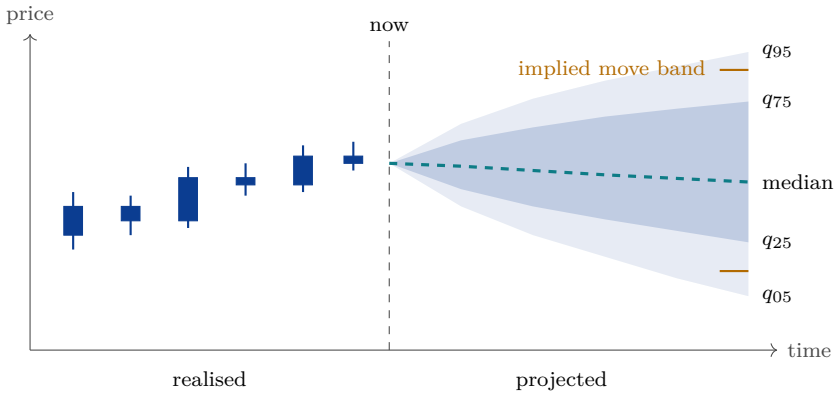
CHAPTER 10

Candle Projection: Honest Ghost Candles

10.1 The redesign

The original design draws a single predicted OHLC sequence. That is scientifically indefensible and, worse, psychologically dangerous: a crisp predicted candle invites the user to bet as if it were certain.

The redesign keeps the visual drama and adds truth: draw a **probability cone** made of quantile candles, where the body and wick widths encode uncertainty and the colour intensity encodes directional confidence.



10.2 How the cone is produced

1. **Volatility forecast** $\hat{\sigma}_h$ from M5 (HAR-RV plus boosted residual), conditioned on the event.
2. **Drift forecast** $\hat{\mu}_h$ from the calibrated direction ensemble converted to an expected return: $\hat{\mu}_h = (2p_{\text{up}} - 1) \cdot \mathbb{E}[|r_h|]$, damped by the meta-labeller's confidence.
3. **Direct quantile heads** from M2 give $q_\alpha(r_h)$ for $\alpha \in \{0.05, 0.25, 0.5, 0.75, 0.95\}$ at each horizon.
4. **Path sampling** for the visual: draw $N = 5000$ paths from a volatility-clustering process (a student-t innovation GARCH-style simulator, or a conditional generative model trained on real intraday paths) calibrated so the terminal distribution matches the quantile heads. Then compute the per-minute empirical quantiles of open, high, low and close.
5. **Rendering**: each projected minute is drawn as a semi-transparent candle whose body spans $[q_{25}, q_{75}]$ of the close and whose wick spans $[q_{05}, q_{95}]$, tinted by $P(\text{up})$.
6. **Overlay the implied move** from the option chain on the same axis. When the cone is narrower than the implied move band, options are expensive relative to our forecast — that is a sell-volatility setup, and it is immediately visible.

The killer visual

Two bands on one chart: *what we forecast* versus *what the market has priced*. The gap between them is the trade. No competing retail product shows this, and it makes the product's thesis legible in one glance.

10.3 Scoring the projection honestly

Metric	Definition	Target
Pinball loss	Quantile loss averaged over α	Must beat a random-walk baseline with GARCH volatility by at least 8% relative.
CRPS	Continuous ranked probability score	Lower than both random walk and the implied-volatility-derived distribution.
PIT histogram	Probability integral transform of realised outcomes	Should be uniform. Humps at the edges mean intervals are too narrow.
Empirical coverage	Fraction of outcomes inside $[q_{05}, q_{95}]$	Within $90\% \pm 3\%$.
Interval width efficiency	Mean width given coverage	Narrower is better only if coverage holds.
Directional deviation	$\lvert\text{realised} - \text{median}\rvert$	Reported, but never as the headline metric.

Reject the temptation

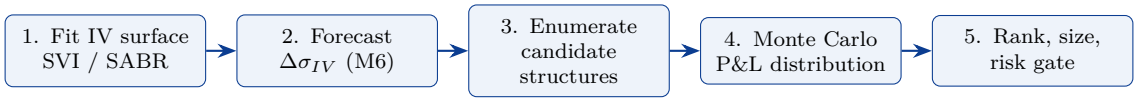
Do not display a single crisp predicted candle even if users ask for it. Instead display the median path as a thin dashed line inside the cone. The moment the product implies certainty, its users will size themselves into ruin and blame the product — correctly.

CHAPTER 11

Options Recommendation Engine

11.1 From a price forecast to a trade

A forecast distribution over the underlying is not a trade. The engine performs five steps.



11.2 Step 1 — The implied volatility surface

Raw per-strike implied volatilities from a live chain are noisy and occasionally arbitrageable due to stale quotes. Fit a smooth, arbitrage-free surface:

- Compute implied volatility from mid prices only (never last traded price — a stale last price generates a fake signal), using Black-76 on futures or Black–Scholes on spot with the appropriate forward.
- Filter: bid > 0, spread below a threshold, open interest above a threshold, and $|\delta| \in [0.05, 0.95]$.
- Fit a parametric slice per expiry (SVI is the standard choice), enforcing no calendar and no butterfly arbitrage.
- Store surface parameters, not just points. Slope, curvature and level become features; the fitted surface prices any strike, including illiquid ones.

11.3 Step 2 — Forecasting the volatility change

The most valuable model in the system

$\Delta\sigma_{IV}$ over the next 15–30 minutes is more forecastable than direction and it dominates short-dated option P&L. Typical patterns to learn:

- Implied volatility *expands* ahead of a scheduled event and *collapses* within minutes of the release — regardless of direction.
- After an unscheduled shock, implied volatility overshoots and mean-reverts with a half-life measured in tens of minutes.
- On expiry day, implied volatility decays mechanically and violently.
- Elevated implied volatility rank means buying premium has a strong headwind.

11.4 Step 3 — Candidate structures

Structure	Use when	Main risk	Notes

Long ATM/OTM call or put	Strong direction, forecast move well above implied, implied volatility rank low	Vega and theta	Simple, highest gamma, but the most exposed to volatility crush.
Vertical debit spread	Direction good, implied volatility elevated	Capped upside	**Default directional structure post-news.** The short leg neutralises much of the vega.
Long straddle / strangle	Big move expected, direction unclear, before a scheduled event	Volatility crush if the move disappoints	Only when forecast move exceeds implied move by a wide margin.
Short strangle with wings (iron condor)	Forecast move well below implied, calm regime	Tail risk	Defined risk only. Never naked short options.
Calendar spread	Term structure dislocated	Complex	Advanced; phase 3 of the options engine.
Ratio / broken-wing	Skew dislocation	Asymmetric	Advanced.
No trade	Forecast within noise of implied	—	Must be the most frequently selected option.

Strike selection is constrained, not free: only strikes with adequate open interest, adequate volume in the last five minutes, and a bid-ask spread below a fraction of the option price are eligible. A theoretically optimal strike that cannot be exited is a trap.

11.5 Step 4 — Monte Carlo P&L distribution

For each candidate structure:

1. Take the N sampled underlying paths from Chapter 10.
2. For each path and each future minute, evolve the implied volatility surface using the M6 forecast plus a stochastic component (correlated negatively with the underlying for equity indices — the leverage effect).
3. Reprice the structure along the path, including the passage of time.
4. Apply the exit rule (target, stop, time stop) to get a realised P&L per path.
5. Subtract full transaction costs on both legs, entry and exit, including modelled slippage as a function of the spread and size.

The output is a P&L distribution, from which we extract:

$$\mathbb{E}[\text{P\&L}], \quad P(\text{profit}), \quad \text{CVaR}_{5\%}, \quad \frac{\mathbb{E}[\text{P\&L}]}{\text{std}}, \quad \text{max drawdown along the path.}$$

Ranking objective

Rank by risk-adjusted expected value after costs, penalised by illiquidity:

$$\text{Score} = \frac{\mathbb{E}[\text{P\&L}] - \lambda\,\text{CVaR}_{5\%}}{\text{capital at risk}} \cdot \underbrace{\left(1 - \frac{\text{spread}}{2\,\text{mid}}\right)^2}_{\text{liquidity penalty}}$$

Emit a recommendation only if $\mathbb{E}[\text{P\&L}] > 0$ after costs *and* the score exceeds a minimum threshold *and* the meta-labeller approves.

11.6 Step 5 — Position sizing

Never size by intuition or by a fixed lot count. Use fractional Kelly with a hard cap:

$$f^* = \frac{p\,b - (1-p)}{b}, \qquad f_{\text{used}} = \min\left(0.25\,f^*,\ f_{\max}\right)$$

where p is the calibrated win probability, b the payoff ratio from the Monte Carlo, 0.25 is the fractional-Kelly discount (full Kelly is far too aggressive when p is estimated), and $f_{\max}$ is a hard per-trade cap (start at 1–2% of capital risked).

Additional size modifier	Effect
Posterior uncertainty of the impact prior	Size scales with $1/\sqrt{1+\text{var}}$; small sample sizes shrink the position.
Regime	Reduce size in high-volatility or low-liquidity regimes.
Time of day	Reduce size in the first 10 and last 15 minutes.
Consecutive losses	Reduce size after a losing streak (anti-martingale).
Correlation with open positions	Aggregate delta and vega across the book, not per trade.
Expiry day	Halve size; gamma risk is extreme.

11.7 Complete recommendation record

Every recommendation is an immutable, fully-explained record. This is both the product's differentiator and the audit trail.

Listing 11.1: Recommendation schema

```
{
  "reco_id": "uuid", "trace_id": "uuid", # links back to the news item
  "ts_emitted": "2026-08-26T09:20:41.318+05:30",
  "trigger": {"event_id": "...", "class": "geopolitical.military_action",
              "severity_cal": 8.2, "novelty": 0.91, "credibility": 0.88,
              "sources": ["src_a","src_b"], "seconds_since_event": 4.1},
  "forecast": {"horizon_min": 15,
               "p_up": 0.31, "p_up_ci": [0.24, 0.39],
               "quantiles": {"q05": -0.0142, "q25": -0.0071, "q50": -0.0033,
                             "q75": 0.0004, "q95": 0.0061},
               "sigma_forecast": 0.0089, "d_iv_forecast": +2.4,
               "implied_move_15m": 0.0052, "edge_ratio": 1.71},
  "structure": {"type": "bear_put_debit_spread",
                "legs": [{"side":"BUY","strike":19500,"opt":"PE","qty":1},
                         {"side":"SELL","strike":19300,"opt":"PE","qty":1}],
                "net_debit": 78.5, "max_profit": 121.5, "max_loss": 78.5,
                "breakeven": 19421.5},
  "greeks_net": {"delta": -0.21, "gamma": 0.0009, "vega": -1.8, "theta": -3.1},
  "mc": {"paths": 5000, "E_pnl": 21.4, "p_profit": 0.57,
         "cvar5": -71.2, "score": 0.184},
  "sizing": {"kelly_frac": 0.061, "used_frac": 0.015, "lots": 2,
```

```
            "capital_at_risk": 11775},
  "exits": {"target": 121.5, "stop": -39.0, "time_stop": "09:36:00",
           "vol_stop": "exit if IV drops > 3 pts"},
  "explanation": ["Military escalation, severity 8.2, first mention 4.1s ago",
                  "Historical prior for this class: median -0.31% over 15m (n=23)",
                  "Forecast |move| 1.71x implied move -> premium buying justified",
                  "IV rank 62 -> spread chosen over naked put to cut vega",
                  "Dealer gamma negative -> moves amplify"],
  "gates_passed": ["credibility>=0.8","two_sources","not_already_priced",
                   "liquidity_ok","daily_loss_limit_ok","regime_ok"],
  "model_versions": {"cls":"ev-cls-v7","dir":"ens-v12","vol":"har-gb-v4",
                     "iv":"ivd-v3","meta":"meta-v5"},
  "mode": "PAPER"
}
```

Why the explanation field is not decoration

Six months from now you will be trying to find out why the system lost money on a class of events. Without a stored, structured reason for every decision, that investigation is impossible and the system cannot improve. The explanation array is a first-class engineering requirement, not a UI nicety.

CHAPTER 12

Risk Management and Kill Switches

12.1 The risk gate sits between the brain and the money

No component may bypass the risk gate. It is deliberately simple, deliberately conservative, and deliberately hard to change (changes require a written justification recorded in the repository).

Limit	Starting value	Behaviour on breach
Per-trade capital at risk	1.5%	Reject or resize down.
Daily loss limit	3%	**Stop for the day.** Flatten everything. No overrides.
Weekly loss limit	6%	Stop for the week; mandatory review before restart.
Max concurrent positions	2	Reject new signals.
Net portfolio delta	bounded band	Reject trades that increase concentration.
Net portfolio vega	bounded band	Prevents accidental all-in volatility bets.
Max trades per day	6	Prevents overtrading in a noisy news day.
Max trades per event	1	Prevents pyramiding on one hypothesis.
Slippage guard	0.5% of premium	Cancel if fill price deviates.
Time-of-day block	first 5 min, last 10 min	No new entries.
Expiry-day multiplier	0.5x size	Gamma risk.
Model degraded flag	any	Signals become advisory only.
Data staleness	feed silent 3 s	Halt everything.

12.2 Kill switch hierarchy

Level	Trigger	Action
L1 Soft	Drift alert, calibration decay, unusual latency	Reduce size by half; notify.
L2 Pause	Daily loss limit, data quality failure, conflicting events	Stop new entries; manage existing positions only.
L3 Flatten	Feed loss, model server failure, clock drift, retraction of the triggering news	Close all positions at market immediately; stop.
L4 Lockout	Two L3 events in a week, or any unexplained loss	System refuses to run until a human clears it with a written root-cause note.

Physical kill switch

Provide a single command and a single dashboard button that flattens everything and disables the system, requiring no dependency on the very components that may have failed. Test it monthly. Also test what happens when the broker API returns an error during the flatten —

that is the scenario that actually costs money.

12.3 Scenario stress tests (run before each promotion)

1. Circuit breaker: the index hits a limit and trading halts mid-position.
2. Flash crash and instant recovery within 90 seconds.
3. Broker API returns success but the order was not placed.
4. Duplicate order acknowledgement (double fill).
5. Option chain goes stale but the spot feed stays live.
6. A false catastrophic headline followed by a retraction 4 minutes later.
7. Two contradictory high-severity events 20 seconds apart.
8. Expiry-day final 30 minutes with implied volatility collapsing.
9. Internet loss for 45 seconds with an open position.
10. Model server returns predictions from a stale model version.

For each: define the expected behaviour, implement it, and write an automated replay test. A system that has not been tested against these has not been tested.

CHAPTER 13

Backtesting and Simulation

13.1 Three levels of simulation

Level	What it is	What it can and cannot tell you
L1 Vectorised	Fast pandas/DuckDB evaluation over historical features and labels	Fast model iteration and feature screening. **Cannot** model execution, latency or queue position. Never trust a P&L number from here.
L2 Event-driven	A simulator that replays ticks, news and chain snapshots in timestamp order through the actual production code path	The real answer. Models latency, partial fills, spread, stale quotes, exits. Slow but honest.
L3 Live paper	The production system running on live data with simulated fills	Catches everything L2 misses: real feed hiccups, real API errors, real timing. **The only evidence that counts for the Phase 7 gate.**

Rule

Model research happens at L1. *Every claim about profitability* must come from L2, and *every go-live decision* must come from L3. A number produced at L1 may never appear in a slide about returns.

13.2 The event-driven simulator

Listing 13.1: Simulator core loop (conceptual)

```
# Single merged, timestamp-ordered stream. This ordering IS the correctness
# guarantee: nothing can be consumed before its t_available.
for msg in merged_stream(ticks, chain_snaps, news, calendar, start, end):
    clock.set(msg.ts)
    bus.publish(msg) # same bus code as production
    for out in engine.drain(): # same engine code as production
        if out.kind == "reco":
            broker_sim.submit(out, latency=sample_latency(),
                              slippage=slippage_model(out, book_state))
    broker_sim.mark_to_market(clock.now())
    risk.check(clock.now())
```

Realism requirement	Implementation
Decision latency	Inject a sampled delay between event timestamp and the moment the engine is allowed to act. Draw from the measured live latency distribution, not a constant.
Fill model	Marketable orders fill at the far touch plus a slippage term that grows with size relative to displayed depth and with spread width. Limit orders fill only if the market trades through them, with a queue-position discount.
Partial fills	Multi-lot orders may partially fill; the engine must handle it.

Stale quotes	If the last chain snapshot is older than x seconds, the simulator refuses the trade, exactly as production must.
Costs	Full stack: brokerage, exchange transaction charge, STT/CTT, SEBI fee, GST, stamp duty, plus modelled slippage. Rates change — keep them in a dated configuration table, applied point-in-time.
Impact	For retail size in index options, market impact is small but not zero; model it as a function of order size relative to trailing volume.
Exits	Simulated with intrabar path assumptions that are *conservative*: if both target and stop are touched within the same minute, assume the stop filled first.

The conservative-tie rule

The stop-first assumption on ambiguous bars typically reduces backtest returns by a large margin. Keep it. Every optimistic assumption you allow is a loan you repay with real money later.

13.3 Cost model

Component	Applies to	Modelling note
Brokerage	Per executed order	Flat per order for most discount brokers; a two-leg spread entered and exited is four orders.
Exchange transaction charge	Premium turnover	Percentage of premium.
Securities transaction tax	Sell side (and exercised options)	Materially different on expiry if allowed to expire in the money — **always square off, never let ITM options expire.**
Regulator fee, stamp duty	Turnover	Small but included.
GST	On brokerage plus charges	Included.
Slippage	Every fill	The dominant real-world cost. Estimate from your own live paper fills, not from theory.

The breakeven reality check

Compute your all-in round-trip cost as a percentage of the option premium at entry. For small premiums and multi-leg structures this can be several percent. The engine must require $\mathbb{E}[\text{P\&L}]$ to exceed this by a comfortable margin — otherwise the strategy is a machine for converting edge into fees.

CHAPTER 14

Evaluation Framework

14.1 Metric hierarchy

Metrics are organised by what decision they inform. Do not mix layers.

Layer	Metrics	Decision it drives
Detection	Precision, recall, macro-F1 per event class; first-mention latency; duplicate rate; false-positive rate on the noise class	Should the classifier ship?
Impact	Coverage of predicted intervals; correlation of predicted and realised abnormal return; sample sizes	Is the impact library trustworthy?
Forecast	Brier score and decomposition; reliability slope and intercept; log loss; AUC; pinball loss; CRPS; PIT uniformity	Is the model calibrated?
Decision	Hit rate, payoff ratio, expectancy per trade, edge realisation (realised versus predicted edge)	Is the decision layer converting forecasts into good trades?
Portfolio	Net P&L after costs, profit factor, Sharpe, Sortino, max drawdown, Calmar, exposure time, turnover, cost drag	Should we scale capital?
Operational	p50/p99 latency per stage, uptime during market hours, data completeness, error budget	Is the system healthy?

14.2 The attribution table — the most useful report you will build

Every trade is tagged with its event class, regime, time bucket, structure type, horizon and model version. The weekly report then answers questions such as:

Event class	n	Hit %	Avg R	Net	Verdict
monetary.rate_decision	12	66.7	+0.81	positive	Keep, size up
macro.inflation_release	19	57.9	+0.34	positive	Keep
geopolitical.rhetoric	31	45.2	−0.22	negative	**Disable this class**

energy.supply_disruption	7	57.1	+0.11	flat	Insufficient sample
flows.expiry_pinning	14	71.4	+0.62	positive	Keep, investigate scaling

Class-level kill policy

Any event class with at least 25 trades and negative expectancy over a rolling quarter is **automatically disabled** by the system, not by a human debate. Re-enabling requires a documented model change and a fresh out-of-sample period. This single policy does more for long-run profitability than any model improvement.

14.3 The public accuracy dashboard, done properly

The original dashboard shows "32/47 correct (68%)". Replace with:

Panel	Content
Reliability curve	Predicted probability bucket versus realised frequency, with confidence bands and the count in each bucket.
Rolling Brier skill score	Versus a climatological baseline; the honest measure of whether the model adds information.
Interval coverage	Realised coverage of the 50% and 90% bands over the trailing 20 sessions.
Edge realisation	Scatter of predicted edge versus realised edge per trade; the regression slope should be near 1. A slope near 0 means the model has no usable edge regardless of hit rate.
Cost drag	Gross versus net P&L, so cost erosion is always visible.
Drawdown curve	Underwater plot with the current position on the risk ladder.
Latency histogram	Per stage, p50/p95/p99, refreshed live.
Per-class attribution	The table above, always visible.

CHAPTER 15

MLOps, Infrastructure and Operations

15.1 Environments

Env	Data	Purpose
`research`	Historical Parquet	Notebooks, feature exploration, model training. No connection to any broker credential, ever.
`replay`	Recorded live days	Deterministic regression testing of the full pipeline. CI runs a fixed golden day on every commit.
`paper`	Live feeds	Production code, simulated fills. Runs every session from Phase 5 onward.
`live`	Live feeds	Real orders. Separate credentials, separate machine, separate alerting, stricter limits.

15.2 The daily operating rhythm

Time (IST)	Job	Detail
06:30	Overnight digest	Collect global overnight events, compute the gap forecast, publish the pre-market briefing.
08:45	Preflight	Feed connectivity, clock sync, model version check, margin check, calendar check (holiday, expiry, special session). Fails loudly.
09:00–09:15	Pre-open watch	Observe only; no signals.
09:15–15:30	Live session	Hot and cold lanes active; risk gate armed.
15:20	Forced flatten	All intraday positions squared off; no exceptions.
15:45	Post-session	Compute outcomes for every prediction, update the ledger, run attribution, push metrics.
18:00	Nightly	Feature materialisation, data-quality gates, archive to Parquet, backup verification.
Weekend	Retrain	Incremental retrain, walk-forward, champion/challenger evaluation, calibration refresh, weekly report generation.

15.3 Observability

- **Trace ID propagation.** One identifier flows from the raw news item through classification, features, prediction, recommendation and fill. Any recommendation can be reconstructed end to end in one query. This is non-negotiable for debugging a system with this many stages.
- **Golden signals per plane:** latency, throughput, error rate, saturation.
- **Business alarms:** no signal emitted for two consecutive sessions (system silently broken), signal rate three times the normal rate (classifier gone haywire), calibration slope outside $[0.8, 1.2]$, feature null rate spike.

- **Every prediction is logged with its feature vector hash** so that post-hoc analysis can recompute exactly what the model saw.

15.4 Reproducibility

Artifact	Discipline
Code	Git; every model artifact records the commit hash.
Data	Immutable, dated Parquet partitions; a dataset version is a manifest of file hashes.
Features	Versioned definitions; changing a feature's logic requires a version bump, never an in-place edit.
Models	MLflow registry with stage transitions (staging, shadow, production, archived). Production reads only from the registry.
Config	All thresholds in versioned YAML, loaded at startup, hash logged. No magic numbers in code.
Randomness	Seeded everywhere; the simulator is deterministic given a seed.

15.5 Security

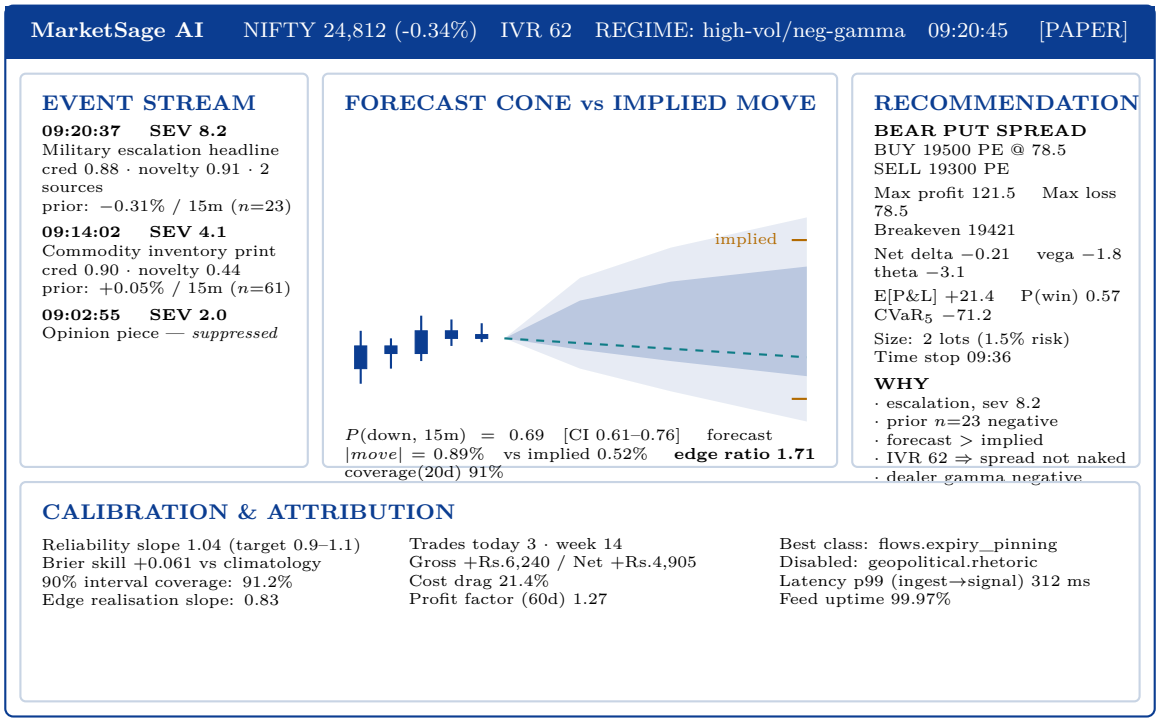
- Broker credentials and API keys in a secrets manager, never in the repository, never in a notebook, never in an environment file committed by accident. Add a pre-commit secret scanner on day one.
- The live trading process runs on a separate host with a minimal attack surface and no inbound ports except an authenticated control channel.
- Two-person rule (or a written checklist plus a delay) for changing risk limits.
- All order-placing code paths require an explicit `live=True` argument that defaults to `False` in every environment.
- Immutable audit log of every order intent, shipped off-host.

CHAPTER 16

Dashboard and Product Surface

16.1 Design brief

The dashboard has one job: let a human decide in three seconds whether to act, and let the same human audit six weeks later why the machine acted. Everything else is decoration.



16.2 Panels and their non-negotiable elements

Panel	Must contain
Event stream	Severity, credibility, novelty, source count, the historical prior *with its sample size*, and suppressed events shown greyed out so the user sees what the system chose to ignore.
Forecast cone	The quantile bands, the median as a thin dashed line, the implied-move band for comparison, the edge ratio, and the trailing coverage statistic so the user knows whether to trust the bands today.
Recommendation	The full structure with both legs, net Greeks, the Monte Carlo summary, the sizing rationale, all three exits (target, stop, time), and the ordered "why" list.
Calibration & attribution	Always visible, never hidden behind a tab. The system's honesty is the product.
Status bar	Mode (PAPER/LIVE/DEGRADED), regime, latency, feed health. Mode must be impossible to misread — a full-width coloured band when live.

Interaction rules

- Nothing flashes or animates except a genuinely new high-severity event.
- Every number is hoverable to reveal its computation and its as-of timestamp.
- Any recommendation can be opened into a full replay: the news item, the feature vector, the model versions, the Monte Carlo distribution.
- A permanent, unmissable indicator of PAPER versus LIVE mode.

16.3 Alerting

Alert	Channel	Rule
Tradeable signal	Push + sound	Only when all gates pass. Rate-limited to avoid alert fatigue.
High-severity event, no trade	Push, silent	"Something is happening, risk is elevated" — valuable even without a trade.
Position exit	Push	Target, stop or time stop hit.
Risk breach	Push + SMS	Any L2/L3 kill switch.
System health	Email/chat	Daily digest; immediate only for L3.
Calibration decay	Daily digest	Slow-moving; does not need to interrupt.

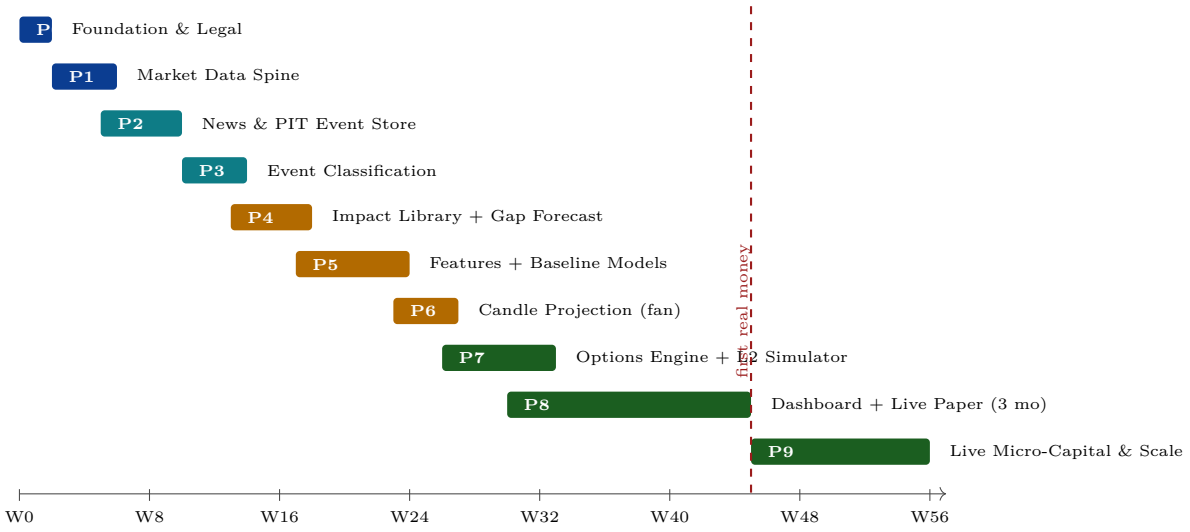
CHAPTER 17

Phase-Gated Execution Roadmap

17.1 How to read this chapter

Ten phases. Each has an objective, workstreams, checkpoints, a numeric **exit gate**, and **kill criteria**. You may not begin phase $n+1$ until phase n's gate is signed off in writing (a dated entry in `docs/gates.md` with the measured numbers pasted in).

The durations assume one committed developer working roughly 25–30 focused hours per week. With a second person on data engineering, compress by around 30%. Do not compress by skipping gates.



17.2 Phase 0 — Foundation and Legal (Weeks 1–2)

Objective. Make it impossible to build the wrong thing, and start the data clock.

CP	Checkpoint	Definition of done
0.1	Repository and environment	Monorepo scaffolded per Chapter 18, Docker Compose brings up Postgres/TimescaleDB, Redis and a service skeleton with one command. Pre-commit hooks including a secret scanner.
0.2	Broker and data vendor decision	Written comparison of at least three brokers on: WebSocket instrument limits, tick frequency, historical data depth, option chain access, algo registration process, order API reliability. Decision recorded with reasons.

0.3	Compliance memo	One page: which regulations apply to personal automated trading through the chosen broker, what data may be stored, what may be shown to others, and what changes if a second user is added. **Written before architecture is frozen.**
0.4	Recorder running	A minimal always-on process writing ticks, option chain snapshots and news headlines with timestamps to disk. Crude is fine. It runs tonight.
0.5	Instrument master	Canonical instrument table, expiry calendar, holiday calendar, lot sizes, tick sizes — all loaded and versioned.

Gate G0

The recorder has produced 5 consecutive complete trading days of ticks, chain snapshots and news with zero unexplained gaps, and the compliance memo is signed. **Every day this gate slips is a day of irrecoverable data.**

17.3 Phase 1 — Market Data Spine (Weeks 3–6)

Objective. A correct, replayable, queryable market history.

CP	Checkpoint	Definition of done
1.1	Tick ingestion service	Reconnect logic, sequence-gap detection, backpressure handling, heartbeat monitoring. Survives a forced disconnect mid-session without data loss.
1.2	Candle builder	1/5/15-minute candles per Chapter 5 rules, including synthetic gap candles and session tagging.
1.3	Option chain snapshotter	Both indices, near two expiries, full strike range, at 1–5 second cadence with completeness metrics.
1.4	TimescaleDB schema + compression + retention	Hypertables, continuous aggregates, compression policy, and a tested restore from backup.
1.5	Parquet archival + DuckDB research layer	A notebook can scan a month of candles in under a second.
1.6	Replay engine	Deterministic replay of a recorded day through the bus. Byte-identical candle output on repeat runs (automated test).
1.7	Data quality gates	The checks in Chapter 5 running nightly with alerts.

Gate G1

(i) Replay a chosen historical day and reproduce candles byte-identically twice. (ii) 20 consecutive sessions captured with $\geq 99.5\%$ expected candles present and option chain completeness $\geq 95\%$. (iii) A full restore from backup succeeds on a clean machine.

Kill criteria for P1

If the chosen broker feed cannot deliver adequate option chain coverage, stop and change vendor now. Discovering this at Phase 7 wastes four months.

17.4 Phase 2 — News Ingestion and Point-in-Time Event Store (Weeks 6–10)

Objective. Build the asset nobody else has: a timestamp-honest archive of what was knowable when.

CP	Checkpoint	Definition of done
2.1	Collector framework	Pluggable per-source collectors, per-source rate limits, retries with backoff, health heartbeats, and a source registry with credibility weights.
2.2	Publish-time resolution	Extract the true publication timestamp; where impossible, set `ts_uncertain` and record the estimation method. Measured accuracy against a manually verified sample.
2.3	Deduplication	Exact hash, SimHash and embedding ANN with cluster heads. Measured duplicate rate.
2.4	Economic calendar with consensus	Scheduled events with expected values loaded ahead of time; surprise computable at release.
2.5	Historical backfill	As much archive as licensing permits, clearly separated from the live-collected archive because their timestamp quality differs.
2.6	Storage and search	PostgreSQL plus vector index; "show me every event of class X in the last two years" returns in seconds.

Gate G2

30 consecutive days collected. Duplicate rate below 3%. On a manually verified sample of 200 items, the publish timestamp is within 5 seconds of truth for at least 90% of items. A point-in-time query ("what did we know at 2026-07-14 11:03:00?") returns a correct, reproducible set.

17.5 Phase 3 — Event Classification (Weeks 10–14)

Objective. Turn text into structured, scored events.

CP	Checkpoint	Definition of done
3.1	Taxonomy v1 frozen	Written definitions with positive and negative examples for every class, including the noise class.

3.2	Gold labelled set	2,000–3,000 items labelled, two annotators on at least 20%, inter-annotator agreement measured and above an agreed threshold. Use an LLM to pre-label and humans to correct — roughly a five-fold speed-up.
3.3	Classifier trained	Fine-tuned encoder, multi-label head, severity regression head, temperature-calibrated confidence.
3.4	Entity and geo extraction	Linked to the canonical knowledge graph, not free text.
3.5	Severity empirical recalibration	Section 6.3 step 3 completed against realised abnormal moves.
3.6	Novelty and confirmation logic	Implemented with the fake-news defence tests from Chapter 6.
3.7	Latency optimisation	Quantised or ONNX-exported; p95 under 120 ms for the tier-2 model.
3.8	Two-tier funnel wired into the bus	End-to-end from raw item to `event.classified`.

Gate G3

Macro-F1 ≥ 0.80 on a temporally held-out test set. Recall on severity-$\geq$7 events ≥ 0.95. False-positive rate on the noise class $\leq 2\%$. Tier-2 p95 latency ≤ 120 ms. The AP-hack injection test produces no directional signal. Severity correlates with realised absolute abnormal return with Spearman $\rho \geq 0.35$.

The severity correlation is the real gate

A classifier can score beautifully on labels and be useless for trading. If severity does not correlate with realised market movement, the taxonomy or the labelling rubric is wrong — fix that before proceeding. This is the most common silent failure in the entire project.

17.6 Phase 4 — Impact Library and the Gap Forecaster (Weeks 13–18)

Objective. Learn what events historically did, and ship the first genuinely useful product.

CP	Checkpoint	Definition of done
4.1	Abnormal-return engine	Time-of-day and regime normalised abnormal returns with matched control sampling.
4.2	Event study over the full archive	Every historical event scored across all horizons; results stored and queryable.
4.3	Hierarchical Bayesian posteriors	Per class and family, with shrinkage, credible intervals and effective sample sizes.
4.4	Volatility and implied-volatility response library	The $\sigma_{\text{post}}/\sigma_{\text{pre}}$ and $\Delta\sigma_{IV}$ distributions per class.
4.5	Cross-asset transfer coefficients	Estimated lead-lag relationships with confidence intervals.

4.6	**Overnight gap forecaster**	A model predicting the next session's opening gap from overnight global events and cross-asset moves — the first component that produces standalone value with no latency requirements.
4.7	Morning briefing product	An automated pre-market note: overnight events, gap forecast distribution, key levels, scheduled events today.

Gate G4

At least 15 event classes have posteriors with $n_{\text{eff}} \geq 8$ and credible intervals excluding zero for at least 5 of them. The gap forecaster beats a naive "gap = overnight global futures move" baseline on out-of-sample CRPS by $\geq 10\%$. The morning briefing has been generated automatically for 20 consecutive sessions.

Why the gap forecaster is placed here

It is the fastest path from "interesting research" to "a thing that is useful every single morning". It needs no low-latency infrastructure, no options engine and no live execution. Shipping it at week 18 rather than at week 45 sustains motivation and produces real feedback — and, being a distributional forecast scored by CRPS, it validates the entire probabilistic approach cheaply.

17.7 Phase 5 — Feature Store and Baseline Prediction (Weeks 17–24)

Objective. Find out whether an intraday edge exists at all.

CP	Checkpoint	Definition of done
5.1	Feature framework	The one-definition/two-materialisation pattern with an automated offline–online parity test in CI.
5.2	Feature groups A, C, E, F implemented	Group B added only if the feed supports it.
5.3	Leakage linter	Automated check of every feature's dependency timestamps against the decision time. Fails the build on violation.
5.4	Triple-barrier labelling	With volatility-scaled barriers, sample uniqueness weights and event grouping.
5.5	Purged/embargoed CV harness	Plus combinatorial purged CV producing a Sharpe *distribution*.
5.6	M0 rules baseline	The bar every model must clear.
5.7	M1 LightGBM direction, M5 volatility, M8 regime	Trained, tuned, SHAP analysed.
5.8	Calibration layer	Isotonic fit plus reliability diagrams.
5.9	M7 meta-labeller	Trained on M1 outputs.
5.10	**Historical option chain purchased**	The hard dependency for Phase 7. Order it at the start of this phase; delivery and cleaning take weeks.

Gate G5 — the decision gate of the whole project

On walk-forward out-of-sample data, restricted to event-conditioned samples and with a 3-second decision delay applied:

- M1 beats M0 on log loss by a statistically significant margin (bootstrap CI excludes zero).
- Calibrated: reliability slope in $[0.85, 1.15]$, Brier skill score > 0.02 versus climatology.
- Directional AUC ≥ 0.55 on high-severity event samples.
- M5 volatility forecast beats a HAR-RV baseline on QLIKE.
- Edge realisation slope > 0.5 in the L1 evaluation.

Kill criteria for P5 — read this twice

If, after honest purged cross-validation with the decision delay applied, AUC sits at 0.50–0.52 and calibration is flat, **there is no intraday directional edge in your current feature set**. Do not proceed to build an options engine on top of a coin flip. Instead: (a) pivot to *volatility-only* strategies, where the gate is far easier to pass; (b) pivot to the overnight gap product from Phase 4, which already works; (c) invest in better data (order book depth, faster sources) and retry once. Three failed attempts at G5 means the honest answer is that this specific edge does not exist at your data tier — and knowing that at week 24 instead of week 80 is a success, not a failure.

17.8 Phase 6 — Distributional Candle Projection (Weeks 23–27)

CP	Checkpoint	Definition of done
6.1	M2 quantile models	Pinball-loss heads at each horizon.
6.2	Conformal wrapper	Split conformal intervals with verified coverage.
6.3	Path sampler	Volatility-clustering simulator calibrated so terminal quantiles match M2.
6.4	Fan-chart renderer	Per-minute quantile candles plus the implied-move overlay.
6.5	Projection scoring	Pinball, CRPS, PIT and coverage tracked daily.
6.6	M3 sequence model (optional)	Only if it beats M1 in the ensemble; drop it without sentiment if it does not.

Gate G6

Out-of-sample 90% interval coverage within $90\% \pm 3\%$; PIT histogram passes a uniformity test; CRPS beats both a random-walk-with-GARCH baseline and the option-implied distribution by $\geq 8\%$ relative. Rendering latency under 120 ms for 5000 paths.

17.9 Phase 7 — Options Engine and Simulator (Weeks 26–33)

CP	Checkpoint	Definition of done

7.1	IV surface fitting	Arbitrage-free parametric fit per expiry with quality metrics and stale-quote filtering.
7.2	Greeks engine	Validated against an independent reference implementation to within tolerance.
7.3	M6 implied-volatility change model	Trained and calibrated.
7.4	Structure enumerator	With liquidity eligibility constraints.
7.5	Monte Carlo P&L engine	Vectorised, under 90 ms p99 for the full candidate set.
7.6	Cost model	Full statutory and brokerage stack, point-in-time rates, plus an empirical slippage model.
7.7	Sizing and risk gate	Fractional Kelly with all modifiers and every limit from Chapter 12.
7.8	L2 event-driven simulator	Production code path, injected latency, realistic fills, conservative tie-breaking.
7.9	Stress-test suite	All ten scenarios from Chapter 12 as automated replay tests.
7.10	**Locked-holdout evaluation**	The single permitted evaluation against the untouched 3-month holdout.

Gate G7

On the L2 simulator over at least 12 months including the locked holdout, and after full costs and a 3-second delay: profit factor ≥ 1.2; positive expectancy in at least 3 of 4 quarters; max drawdown $\leq 12\%$ of notional risk capital; deflated Sharpe ratio > 0; and the strategy survives all ten stress scenarios with the specified behaviour. Every recommendation carries a complete explanation record.

17.10 Phase 8 — Dashboard and Three Months of Live Paper Trading (Weeks 30–45)

Objective. Prove it in the wild, with no money at risk. This phase cannot be compressed — it is bounded by calendar time, not by effort.

CP	Checkpoint	Definition of done
8.1	Dashboard v1	All five panels from Chapter 16, WebSocket live, with the PAPER/LIVE indicator.
8.2	Alerting	Push notifications with rate limiting.
8.3	Paper ledger	Every signal recorded with simulated fills at live prices, plus outcome resolution at horizon.
8.4	Daily operating rhythm automated	Preflight, session, post-session, nightly and weekend jobs all running unattended.
8.5	Observability	Trace propagation, dashboards, alarms, error budget.
8.6	**60 trading sessions of live paper**	Unattended, with an availability target of 99% during market hours.
8.7	Weekly attribution reviews	12 written reviews with actions taken.

8.8	Class-level kill policy active	Underperforming event classes automatically disabled.

Gate G8 — the go-live gate

Over ≥ 60 sessions and ≥ 150 paper trades:

- Positive net expectancy after full modelled costs.
- Profit factor ≥ 1.15.
- Max drawdown $\leq 10\%$ of notional risk capital.
- Reliability slope in $[0.85, 1.15]$ on live data.
- Live results within one standard error of the L2 simulator's prediction for the same period (**if live and simulated diverge, the simulator is wrong and everything built on it is suspect**).
- Zero unhandled exceptions in the trading path; every kill switch exercised at least once in a drill.
- Broker algo registration complete; compliance memo updated.

17.11 Phase 9 — Live Micro-Capital and Scaling (Week 45 onward)

Rung	Capital	Promotion criteria to the next rung
R1	Minimum viable (1 lot, smallest instrument)	20 sessions, live P&L within one standard error of paper P&L for the same signals, slippage within the modelled band.
R2	2x R1	40 sessions, profit factor ≥ 1.15, no L3 kill events.
R3	4x R1	60 sessions, Sharpe > 1 annualised, drawdown within limits.
R4	8x R1	Two consecutive profitable quarters; slippage still within model.

Demotion rule: any breach of the weekly loss limit, or live results deviating from paper by more than two standard errors, drops the ladder by one rung immediately and triggers a written review. Scaling up is slow; scaling down is instant.

The paper-versus-live divergence test

Run the paper engine *alongside* the live engine forever, on identical signals. The difference between paper and live P&L is your true execution cost. If that gap widens, the model has not degraded — your fills have. These are different problems with different fixes, and without running both you cannot tell them apart.

17.12 Phase summary table

Ph	Name	Weeks	Single most important output
P0	Foundation and Legal	1–2	The recorder is running tonight.
P1	Market Data Spine	3–6	Deterministic replay.

P2	News and PIT Event Store	6–10	Timestamp-honest archive.
P3	Event Classification	10–14	Severity that correlates with real moves.
P4	Impact Library + Gap Forecaster	13–18	Conditional base rates, and the first shipped product.
P5	Features + Baseline Models	17–24	The honest answer to "is there an edge?"
P6	Candle Projection	23–27	Calibrated forecast cones.
P7	Options Engine + Simulator	26–33	Trades whose expected value is positive after costs.
P8	Dashboard + Live Paper	30–45	60 sessions of unattended live evidence.
P9	Live Capital	45+	A capital ladder that only climbs on evidence.

17.13 The first fourteen days, concretely

Day	Task
1	Create the repository and the Docker Compose stack (Postgres/TimescaleDB, Redis). Get `docker compose up` working. Add pre-commit with a secret scanner.
2	Open the broker account with API access; obtain and store credentials in a secrets manager. Read the API documentation end to end and write down the instrument subscription limit and tick frequency.
3	Write the tick recorder. Subscribe to the two index spots and futures. Write raw messages to disk as newline-delimited JSON. Ugly is fine.
4	Write the option chain snapshotter. Both indices, near two expiries. Run it for a full session and measure completeness.
5	Write two news collectors (one wire RSS, one official/central-bank page). Store raw with three timestamps.
6	Build the instrument master, expiry calendar and holiday calendar. Load them.
7	Review the week: verify that five days of data exist and are complete. Fix gaps. **This recorder never stops again.**
8	TimescaleDB schema; a loader from raw files into hypertables.
9	The candle builder with the six correctness rules; write the determinism test.
10	Parquet archival job and a DuckDB notebook; plot a day of Nifty from Parquet.
11	Data quality checks: candle counts, tick gaps, chain completeness. Wire an alert.
12	The compliance memo. Talk to the broker about algo registration requirements. Write it down.
13	The broker/vendor comparison document; commit to a decision.
14	Gate G0 review. Write the numbers into `docs/gates.md`. Only then start Phase 1 properly.

What not to do in the first fourteen days

Do not train a model. Do not build a dashboard. Do not integrate an LLM. Do not optimise latency. Every one of those is more fun than building a recorder, and every one of them is

worthless without the data the recorder produces.

CHAPTER 18

Repository Structure and Engineering Conventions

Listing 18.1: Monorepo layout

```
marketsage/
  docs/
    marketsage_master_plan.tex # this document
    gates.md # SIGNED phase-gate records with numbers
    decisions/ADR-0001-broker.md # architecture decision records
    compliance_memo.md
  config/
    instruments.yaml costs.yaml # dated, point-in-time cost tables
    risk_limits.yaml # changes require an ADR
    sources.yaml # source registry + credibility weights
    taxonomy.yaml # versioned event taxonomy
  libs/
    ms_core/ # schemas, clock, bus client, tracing, config loader
    ms_data/ # tick handler, candle builder, chain snapshotter, QC
    ms_news/ # collectors, normaliser, dedup, publish-time resolver
    ms_events/ # classifier, severity, novelty, confirmation, KG routing
    ms_impact/ # event study, abnormal returns, Bayesian posteriors
    ms_features/ # feature registry, offline+online materialisers, linter
    ms_models/ # training, CV harness, calibration, conformal, registry
    ms_options/ # IV surface, greeks, structures, Monte Carlo, sizing
    ms_risk/ # limits, kill switches, portfolio state
    ms_sim/ # L2 event-driven simulator, fill + slippage models
    ms_broker/ # broker adapters (paper + live), order state machine
  services/
    ingest_market/ ingest_news/ engine_event/ engine_predict/
    engine_decide/ api_gateway/ ui_web/
  jobs/
    nightly_features.py weekly_retrain.py event_study_rebuild.py
    postsession_outcomes.py morning_briefing.py
  tests/
    unit/ replay/golden_day_2026_07_14/ stress/ parity/
  notebooks/ # research only; never imported by services
```

Convention	Rule
No magic numbers	Every threshold lives in `config/`, is loaded at startup, and its hash is logged with each decision.
Schemas first	Every bus message has a versioned schema; consumers validate. Adding a field is allowed, changing a meaning requires a version bump.
Notebooks are disposable	A notebook may never be imported by a service. If code matters, it moves into a library with tests.
Time is explicit	No implicit `now()` in library code — the clock is injected, so the simulator and production share the same code path.
Live requires intent	Any function that can place a real order takes `live: bool = False` and logs loudly.
Tests that matter	Determinism test, offline–online parity test, leakage linter, golden-day replay, stress scenarios. These run in CI on every commit.
Gate discipline	`docs/gates.md` is the project's spine. No gate entry, no phase advance.

CHAPTER 19

Resources, Cost and Risk Register

19.1 Team shapes

Shape	Timeline to G8	Notes
Solo (you)	45–60 weeks	Entirely feasible. The binding constraint is data-engineering grind, not modelling. Expect Phases 1–2 to feel slow and unglamorous; they are the moat.
Two people	30–38 weeks	Optimal split: one owns data and infrastructure (P0–P2, P7 simulator, P8 ops), the other owns ML and research (P3–P6).
Three to four	24–30 weeks	Add a dedicated frontend and a research generalist. Beyond four, coordination cost exceeds benefit at this stage.

19.2 Cost structure (relative, not absolute)

Prices change; the *ratios* do not. Plan the budget in these proportions.

Item	Share of year-1 spend	Comment
Historical option chain data	**Largest single item**	Unavoidable for a credible options backtest. Buy once, clean carefully, treat as an asset.
Live market data (broker or vendor)	Moderate, recurring	Start on the broker feed; upgrade only when the gate demands it.
News and social API tiers	Moderate, recurring	Start with 3–5 high-value sources rather than breadth.
Compute (cloud VM in the correct region, storage)	Moderate, recurring	One solid VM plus fast disk beats a Kubernetes cluster at this stage.
GPU for training	Small	Rent by the hour for training runs; do not buy or keep one running. Inference runs on CPU via ONNX.
Trading capital at R1	Deliberately tiny	Capital is the *last* thing you spend, not the first.
Legal/compliance consultation	Small, one-off	Cheaper than one regulatory problem.

Budget principle

Spend on **data** first, **time** second, **compute** third and **trading capital** last. Most failed projects invert this exactly.

19.3 Risk register

Risk	Prob	Impact	Mitigation
No intraday directional edge exists at our data tier	High	High	Gate G5 tests this at week 24 for a fraction of the total cost. Pivot options are pre-defined: volatility-only, or the gap product.
Historical option chain data unavailable or unaffordable	Med	High	Identify vendors and prices in Phase 0, not Phase 5. If truly blocked, restrict v1 to futures and the gap product.
Backtest overfitting	**Very high**	**Very high**	Purged CV, embargo, combinatorial CV, deflated Sharpe, locked holdout evaluated once, experiment log. This is the default outcome unless actively fought.
Live results diverge from paper	High	High	Run paper alongside live permanently; the gap is measured, not guessed. Capital ladder demotes instantly.
Slippage larger than modelled	High	Med	Calibrate the slippage model from your own fills; require expected value to clear costs with margin.
Fake or retracted news triggering a trade	Med	**Very high**	Multi-source confirmation, severity gating, retraction handling, the AP-hack CI test.
Regime change invalidating models	High	Med	Regime-conditional models, drift monitors, automatic class disabling, scheduled retrains.
Broker API outage or rejection mid-position	Med	High	Order state machine with reconciliation, flatten-on-failure drills, a manual override runbook.
Regulatory change affecting retail algo access	Med	High	Compliance memo maintained; architecture keeps the human-approval mode as a first-class path, not a hack.
Data licensing violation when adding users	Med	High	Legal review before Tier T5. Do not show live exchange prices to third parties without a licence.
Solo-developer burnout	**High**	High	Phase 4 ships a real product at week 18. Weekly gates give visible progress. The plan is explicitly designed so that stopping at any gate still leaves something valuable.
Scope creep (adding stocks, crypto, more countries)	Very high	Med	Chapter 2 scope decisions are frozen until G8. Write new ideas in a backlog file; do not build them.

19.4 What "done" looks like at each tier, restated as artefacts

Tier	The artefact that proves it
T0	A year of complete, replayable, point-in-time data on disk with automated quality reports.
T1	An automated daily note explaining yesterday's moves, rated by you against reality for 20 sessions.
T2	A reliability diagram from live paper trading with a slope near 1 and a positive Brier skill score.

T3	A signed G8 record: 60 sessions, 150 trades, positive expectancy after costs, live matching simulated.
T4	A broker statement showing real fills on ladder rung R1, and a paper-versus-live divergence chart within tolerance.
T5	A legal opinion, a data licence, a multi-tenant deployment and an SLO report.

APPENDIX A

Reference Formulas

Black–Scholes / Black-76 Greeks

With $d_1 = \dfrac{\ln(F/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$, $d_2 = d_1 - \sigma\sqrt{T}$, discount factor $D = e^{-rT}$:

$$C = D\big[F\,N(d_1) - K\,N(d_2)\big], \qquad P = D\big[K\,N(-d_2) - F\,N(-d_1)\big],$$

$$\Delta_C = D\,N(d_1), \qquad \Gamma = \frac{D\,\phi(d_1)}{F\sigma\sqrt{T}},$$

$$\mathcal{V} = D\,F\,\phi(d_1)\sqrt{T}, \qquad \Theta_C = -\frac{DF\phi(d_1)\sigma}{2\sqrt{T}} - rDKN(d_2) + rDFN(d_1).$$

Intraday scaling of implied volatility

$$\sigma_{h\text{ minutes}} = \sigma_{\text{annual}}\sqrt{\frac{h}{375 \times 252}}, \qquad \text{Implied move}_h = S\,\sigma_h.$$

Range-based realised volatility (Garman–Klass)

$$\sigma^2_{GK} = \tfrac{1}{2}\big(\ln\tfrac{H}{L}\big)^2 - (2\ln 2 - 1)\big(\ln\tfrac{C}{O}\big)^2.$$

HAR-RV volatility model

$$RV_{t+1} = \beta_0 + \beta_d RV_t^{(d)} + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+1},$$

with daily, weekly and monthly averages of realised variance. Add a gradient-boosted model on the residual with event features.

Order-flow imbalance

$$OFI_t = \sum_{i \in \text{window}} \big[\mathbb{1}_{\{P_i^b \geq P_{i-1}^b\}} q_i^b - \mathbb{1}_{\{P_i^b \leq P_{i-1}^b\}} q_{i-1}^b - \mathbb{1}_{\{P_i^a \leq P_{i-1}^a\}} q_i^a + \mathbb{1}_{\{P_i^a \geq P_{i-1}^a\}} q_{i-1}^a\big].$$

Pinball (quantile) loss

$$L_\alpha(y, \hat{q}_\alpha) = \begin{cases} \alpha\,(y - \hat{q}_\alpha) & y \geq \hat{q}_\alpha \\ (1-\alpha)(\hat{q}_\alpha - y) & y < \hat{q}_\alpha \end{cases}$$

Continuous ranked probability score

$$\text{CRPS}(F, y) = \int_{-\infty}^{\infty} \big(F(x) - \mathbb{1}_{\{x \geq y\}}\big)^2 dx \;\approx\; \frac{1}{N}\sum_i |x_i - y| - \frac{1}{2N^2}\sum_{i,j} |x_i - x_j|$$

for N sampled paths x_i — directly computable from the Monte Carlo output.

Brier score decomposition

$$BS = \underbrace{\text{Reliability}}_{\text{lower is better}} - \underbrace{\text{Resolution}}_{\text{higher is better}} + \underbrace{\text{Uncertainty}}_{\text{fixed by the data}}$$

Fractional Kelly

$$f^* = \frac{pb - (1-p)}{b}, \qquad f_{\text{used}} = \min(0.25 f^*, f_{\max}).$$

Deflated Sharpe ratio (concept)

Given N independent strategy trials, the expected maximum Sharpe under the null grows roughly with $\sqrt{2 \ln N}$. Always report how many variants were tested; an unreported trial count makes any Sharpe ratio meaningless.

APPENDIX B

Event Taxonomy Skeleton (YAML)

Listing B.1: config/taxonomy.yaml — abridged

```yaml
version: 1
families:
  monetary:
    classes:
      rate_decision: {sev_range: [6,10], scheduled: true, horizons: [5,15,30,60]}
      stance_change: {sev_range: [5,9], scheduled: true}
      liquidity_measure: {sev_range: [4,8], scheduled: false}
      unscheduled_action: {sev_range: [8,10], scheduled: false}
  macro:
    classes:
      inflation_release: {sev_range: [4,8], scheduled: true, needs_consensus: true}
      growth_release: {sev_range: [4,8], scheduled: true, needs_consensus: true}
      trade_balance: {sev_range: [3,6], scheduled: true, needs_consensus: true}
  geopolitical:
    classes:
      military_action: {sev_range: [6,10], scheduled: false, require_sources: 2}
      escalation_rhetoric:{sev_range: [3,7], scheduled: false, require_sources: 2}
      sanctions: {sev_range: [4,8], scheduled: false}
  energy:
    classes:
      production_quota: {sev_range: [5,9], scheduled: true}
      supply_disruption: {sev_range: [4,9], scheduled: false}
  flows:
    classes:
      expiry_pinning: {sev_range: [3,6], scheduled: true, intraday_only: true}
      oi_shift: {sev_range: [3,6], scheduled: false}
  noise:
    classes:
      opinion: {sev_range: [0,2], tradeable: false}
      repetition: {sev_range: [0,1], tradeable: false}

gates: # applied before any directional signal
  min_severity: 5.0
  min_credibility: 0.75
  min_novelty: 0.45
  max_already_priced: 0.6
  min_effective_sample: 8
```

APPENDIX C

Glossary

Term	Meaning in this document
Abnormal return	Return minus what was expected given time of day, regime and day type. The correct unit for event studies.
Calibration	Whether stated probabilities match realised frequencies.
Conformal prediction	A method producing prediction intervals with distribution-free finite-sample coverage guarantees.
CRPS	A proper scoring rule for a full predictive distribution.
Dealer gamma exposure	Aggregate options-dealer gamma; when negative, hedging amplifies moves.
Edge realisation	The regression slope of realised edge on predicted edge. A slope near zero means no usable edge, regardless of hit rate.
Embargo	A buffer period dropped after each test fold to prevent leakage from serial correlation.
Implied move	The move the option market has already priced for a horizon.
IV crush	A fall in implied volatility that can make a directionally correct option trade lose money.
Meta-labelling	A second model that decides whether to act on the first model's signal.
Novelty	Whether an item carries genuinely new information not already priced.
Point-in-time	Data as it was actually knowable at a past moment, including timestamps and later revisions.
Purged CV	Cross-validation that removes training samples whose outcome windows overlap the test window.
Regime	A market state (volatility, trend, liquidity) that conditions all other models.
Triple barrier	Labelling by which of the upper, lower or time barrier is hit first.
Volatility risk premium	Implied volatility minus forecast realised volatility; the core edge for volatility structures.

APPENDIX D

Final Word

The three sentences that matter

1. **Start the recorder tonight.** Models can be rebuilt; history cannot be recovered.
2. **Gate G5 at week 24 is the real project.** Everything before it exists to make that test honest; everything after it exists to exploit a positive answer safely.
3. **Calibration, not accuracy, is the product.** A system that knows what it does not know is the only kind that survives contact with a live market.

This blueprint is deliberately harder than the original concept, because the original concept was designed to impress and this one is designed to survive. The vision is unchanged: watch the world, understand the impact, forecast the distribution, recommend the trade, and score yourself in public. The difference is that every claim here has a test attached to it, and every test has a date.